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Abstract

In recent years, diffusion models have advanced significantly, resulting in specialized models trained for specific objectives, such as generating content with unique styles and identities. However, many tasks remain challenging to address in real-world scenarios like dancing video generation, since training models for these applications requires substantial computational resources and large-scale datasets. To overcome these limitations, various model-fusion strategies have been proposed to combine the strengths of different models. Although these methods have shown some success, they are typically restricted to homogeneous architectures and uniform training paradigms, limiting their applicability across diverse model types.

This thesis presents **AdaptiveFuse**, a novel noise-level fusion framework designed to seamlessly integrate diffusion models with diverse architectures, training paradigms, and sampling schedules without additional training. Unlike existing methods that rely on shared backbones or retraining models, AdaptiveFuse dynamically combines the noise residual predictions of each model at every denoising step. Moreover, we introduce adjustable fusion parameters and adaptive scheduling mechanisms, allowing fine-grained, dynamic control over each model's contribution to specific attributes such as style consistency or facial detail preservation.

To demonstrate AdaptiveFuse's flexibility and effectiveness, we apply it across multiple challenging scenarios: character image stylization by fusing models enhanced with Character LoRA and Style LoRA, dancing video generation by integrating a face-centric image-generation model with a motion-specialized video-generation model, and character video stylization incorporating camera motion through Diffusion Transformer models. These experiments show that AdaptiveFuse achieves superior visual fidelity, and improved identity preservation compared to existing methods. By enabling fusion across heterogeneous models, AdaptiveFuse broadens the applicability of generative modeling. Ultimately, our approach facilitates tackling complex, resource-intensive generative tasks, bridging the gap between specialized models and practical, multi-objective real-world applications.

Keywords: Diffusion Models, Model Fusion, Video Generation, Identity Preservation.

摘要

近年来，扩散模型取得了显著的发展，涌现出大量专为特定任务而训练的模型，如生成具备独特风格与身份特征的内容。然而，许多实际应用场景仍存在挑战，例如舞蹈视频生成等任务，因其对计算资源与大规模数据集的需求较高，模型训练难度较大。为克服这些限制，各种模型融合策略被提出以结合不同模型的优势。尽管这些方法取得了一定成功，但通常局限于架构相似、训练范式统一的模型之间，难以应用于多样化的模型类型。

本论文提出了一种名为 **AdaptiveFuse** 的新型噪声级融合框架，能够在无需额外训练的情况下，无缝地融合具备不同架构、训练范式和采样过程的扩散模型。与传统方法依赖于共享架构或模型重训不同，AdaptiveFuse 在去噪过程的每一步中动态融合每个模型预测的噪声残差。此外，我们引入了可调的融合参数与自适应调度机制，使用户能够对每个模型在特定属性（例如风格一致性或面部细节保留）上的贡献进行精细且动态的控制。

为了展示 AdaptiveFuse 的灵活性与有效性，我们在多个具有挑战性的场景中进行了验证，包括：融合分别配备角色 LoRA 与风格 LoRA 的模型进行角色图像风格化；融合一个专注于面部细节的图像生成模型与一个专注于动作生成的视频生成模型进行舞蹈视频生成；以及通过扩散 Transformer 模型融合包含摄像机运动的角色视频风格化模型。这些实验表明，AdaptiveFuse 在视觉真实度与身份特征保留方面均显著优于现有方法。通过实现异构模型之间的有效融合，AdaptiveFuse 拓宽了生成模型的适用范围，最终帮助解决复杂且资源密集型的生成任务，弥合了专用模型与实际多目标应用需求之间的差距。

关键词：扩散模型、模型融合、视频生成、身份保真

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List of Symbols

Symbol	Description
CLIP	Contrastive Language-Image Pre-training
DDIM	Denoising Diffusion Implicit Models
DDPM	Denoising Diffusion Probabilistic Models
FID	Fréchet Inception Distance (image quality metric)
FVD	Fréchet Video Distance (video consistency metric)
GAN	Generative Adversarial Network
LDM	Latent Diffusion Models
LoRA	Low-Rank Adaptation (efficient fine-tuning method)
LPIPS	Learned Perceptual Image Patch Similarity (perceptual metric)
PSNR	Peak Signal to Noise Ratio (image fidelity metric)
SSIM	Structural Similarity Index Measure (image fidelity metric)
VAE	Variational Autoencoder



Chapter 1. Introduction

1.1 Research Background

Over the past decade, generative modeling has seen significant advancements, in part due to methods such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and most recently, diffusion models. Among these methods, **diffusion models** have gained remarkable traction due to their robust training stability, broad coverage of target data distributions, and capacity to generate high-fidelity images and videos. Initially established by foundational studies [1–3], diffusion models operate by incrementally adding noise to training samples and subsequently learning to reverse this corruption process in multiple steps. This multi-step denoising paradigm mitigates problems such as mode collapse (common in GANs) and allows more stable training. As a result, diffusion-based approaches now underpin a wide array of tasks, ranging from conditional image generation [4, 5] to high-resolution video synthesis [6, 7].

Despite this rapid progress, real-world applications rarely involve a single, uniform generative goal. Instead, practitioners often require *multiple specialized functionalities*, such as synthesizing diverse visual styles, preserving specific subject identities, or generating complex temporal motions (e.g., dancing videos). A common approach is to fine-tune an existing, pre-trained diffusion model on a specialized dataset geared toward stylization, personalization, or advanced motion generation [8–10]. However, training or fine-tuning a *single* diffusion model to meet multiple disparate objectives simultaneously remains difficult. Mixing data from heterogeneous tasks can create conflicts, cause catastrophic forgetting, or degrade performance unless large-scale re-training is performed.

In response, recent studies have focused on methods to *fuse multiple diffusion models* (or specialized modules) into a unified generative pipeline, consolidating a range of abilities. Early work on model fusion often emphasized networks with *homogeneous architectures* and consistent training schedules, since direct weight-based merges can work when models share nearly



identical backbones [11]. More recently, a broader taxonomy of fusion strategies has emerged, including:

1. **Weight-Based Fusion:** Merging the parameter updates (for instance, LoRA weights) from different tasks into a single model [11]. As one example, *ZipLoRA* [11] fuses multiple LoRA modules trained on different styles or subjects by carefully adjusting merged columns to avoid destructive interference.
2. **Switch-Based Fusion:** Switching between specialized model segments or LoRA layers at different timesteps [12]. For example, *Multi-LoRA Composition* [12] activates distinct modules at specific stages to blend various styles or subjects.
3. **Noise-Based Fusion:** Combining the *noise residuals* predicted by different diffusion models to incorporate complementary capabilities. Notable examples include *Explorative Inbetweening of Time and Space* [13] and *NVS-Solver* [14], which manipulate noise-space trajectories to merge pre-trained generative models while preserving each model’s individual strengths.

However, many existing fusion methods assume that all models share compatible architectures or rely on uniform sampling schedules. In practice, a user may want to integrate, for example, a latent diffusion model that preserves identity with a motion-focused video diffusion model that employs a different noise schedule or an additional motion encoder. Simply merging weights or switching modules can fail when the underlying diffusion processes diverge significantly.

Our Motivation and Proposed Approach. In this thesis, we propose AdaptiveFuse, a noise-fusion paradigm that is model-agnostic, uniquely enabling the integration of *heterogeneous diffusion models* trained with different sampling schedules or distinct architectures. Unlike existing approaches that typically require re-training or enforcing a shared backbone, our method dynamically fuses each model’s predicted noise residuals at every denoising step. This noise-level fusion is architecture-agnostic, because diffusion models produce noise predictions with



identical dimensions and semantic meaning in the latent space, allowing seamless integration of models with different architectures and training paradigms.

Crucially, our approach introduces adjustable fusion parameters, allowing precise temporal control over each model’s contribution throughout the denoising process. This capability facilitates fine-grained manipulation of complex attributes, such as style consistency or detailed facial features. Furthermore, our fusion paradigm supports both *learned and user-defined schedules*, providing flexibility to adjust subtle characteristics that pose significant challenges for standalone diffusion models, such as accurately preserving facial details during dynamic activities like dancing in videos.

To showcase the flexibility of this framework, we apply noise-fusion across diverse architectures. First, we address image stylization by fusing two image-generation models enhanced separately with Character LoRA and Style LoRA. Second, we tackle dancing video generation by integrating models with distinct strengths—one image-generation-based model excelling in facial rendering, and another video-generation model specialized in dance video synthesis. Third, we experiment with the Diffusion Transformer model by fusing models enhanced individually with LoRAs focused on character and camera motion, and another with Style LoRA. Our work extends beyond existing studies, which primarily fuse only image-generation models, by also merging entirely different architectures such as dance video-generation models and the promising Diffusion Transformer model, demonstrating the broad applicability of AdaptiveFuse across multiple domains. Our experiments demonstrate that dynamic noise-level fusion provides superior temporal consistency, facial fidelity, and overall quality compared to straightforward alternatives, weight-space merging or switching.

1.2 Related Work

We give a concise overview of each research direction, referring to relevant studies and explaining how they motivate a training-free fusion solution.



1.2.1 Fundamentals of Diffusion Models

Early efforts in generative modeling, including Variational Autoencoders (VAEs) [15], Generative Adversarial Networks (GANs) [16], and Normalizing Flows [17], established different paradigms for mapping high-dimensional data manifolds. Yet, each faced limitations, such as training instability or scalability issues. Diffusion models [1–3] address these challenges by incrementally adding noise in a forward process and learning to remove it in a backward process. Compared with GANs, this multi-step approach often provides better training stability and more comprehensive coverage of target distributions. Subsequent developments (score-based models [3, 18], DDPMs [2], and LDMs [5]) have optimized sampling efficiency and enabled high-resolution synthesis. Still, fusing *multiple specialized diffusion models* remains an open problem [14].

1.2.2 Diffusion for Image and Video Generation

High-Fidelity Image Generation. Diffusion models have become particularly prominent for image-related tasks [4, 5], often surpassing methods like GAN in both realism and variety. Moreover, their iterative structure supports *controllable* outputs via conditioning signals such as text [19], keypoints [20], or LoRA modules [8, 11]. Approaches like *Stable Diffusion* [5], *ControlNet* [20], and *IP-Adapter* [21] broaden the applicability of diffusion pipelines to tasks including style transfer [9], identity preservation [22], or multi-conditioned generation.

Video Generation. Building on progress in image domains, diffusion models have also been customized for video synthesis, with a focus on spatiotemporal consistency and realistic motion [6, 7]. Recent work employs specialized attention modules or motion-centric components, such as *AnimateDiff* [7], to maintain multi-frame coherence. However, many video diffusion models struggle with retaining subject identity or detailed facial features, especially in tasks like dance-oriented animations [10, 23]. These issues highlight the potential of combining motion specialization with other targeted features (e.g., identity or style modules) in a single framework.

1.2.3 Model Fusion Techniques

Since training one diffusion model for multiple goals can be difficult, *model fusion* has gained traction as a way to exploit pre-trained modules [14]. Existing methods generally presume similar architectures and noise schedules, and they mainly fall into three groups:

Weight-based fusion merges specialized knowledge by summing parameter updates (e.g., LoRA weights) from different tasks into one base model. *ZipLoRA* [11] is a notable example, combining LoRA modules fine-tuned on various styles or subjects while adjusting merged columns to reduce interference. Though this method can be effective, it relies on uniform network structures or training protocols and can lead to destructive conflicts when tasks differ substantially [11].

Switch-based fusion activates separate specialized modules at distinct timesteps during the denoising process. *LoRA Composite* [12], for instance, swaps in different LoRA modules at various points to achieve multi-style or multi-subject outputs in a single inference pass. However, switch-based methods may not suffice when *fine-grained* fusion is needed, for example merging partial features from two models in the same frame.

Noise-based fusion differs by mixing the noise residuals from multiple models, instead of merging their weights. Recent works, such as *NVS-Solver* [14] and *Explorative Inbetweening* [13], adjust the denoising path in noise space, avoiding direct parameter conflicts. Though these techniques are promising, they often require uniform architectures or sampling schedules, which reduces their flexibility in scenarios where the model designs or sampling approaches vary.

These restrictions pose challenges for *multi-objective* tasks, such as style transfer, identity preservation, or advanced motion generation, especially if the models involved have different training or architectural assumptions [11, 12]. Therefore, a more flexible, training-free integration at the noise level is in high demand when dealing with heterogeneous diffusion models (e.g., merging an image-focused model with a video-focused one).



1.2.4 Dancing Video Generation

Dancing video generation exemplifies the difficulties of merging different specialized capabilities. The model must produce both plausible body motion following dance poses [24, 25] and consistent subject identity (facial features, expressions, and other distinguishing traits) across many frames [10]. Systems like *DisCo* [24], *MagicAnimate* [23], and *AnimateAnyone* [10] employ different modules for pose, appearance, or facial refinement, but they often involve significant re-training or a single unified architecture, making it cumbersome to add or swap in a new motion model or face-detail module.

Noise-based fusion stands out as a promising avenue: by carefully blending noise predictions from a motion-centric and a face-centric model, one can preserve each model’s strengths [14]. Nevertheless, handling different noise schedules, training conditions, and model architectures remains non-trivial. This challenge drives the need for a *dynamic*, model-agnostic fusion method, which is the central idea behind our approach.

1.2.5 Diffusion Transformer Model

Text-to-video generation models represent a significant advancement in generative AI, allowing users to create dynamic visual content from textual descriptions. Recent developments have been primarily driven by two architectural paradigms: diffusion models with convolutional U-Nets and diffusion transformers (DiTs). Early text-to-video approaches like CogVideo [26] and Phenaki [27] demonstrated that large-scale pre-trained transformers could generate short videos from text prompts, albeit with limited quality and temporal consistency. A breakthrough came with Make-A-Video [28] and Imagen Video [6], which applied diffusion-based techniques to video generation, significantly improving visual fidelity and motion coherence. More recently, transformer-based diffusion models have pushed the boundaries of text-to-video generation. CogVideoX [29] employs a 3D causal VAE and expert transformer architecture to generate videos with improved temporal coherence and semantic alignment. This model uses specialized components like Expert Adaptive LayerNorm to better integrate text and visual information, while its 3D full attention mechanism captures complex spatial-temporal relationships more



effectively than separated attention approaches. Despite these advances, text-to-video models still face significant challenges. First, they require massive computational resources for training and inference, limiting widespread deployment. Second, they often struggle with precise control over specific visual attributes such as camera movements, stylistic elements, and character identity simultaneously. Third, maintaining consistent subject identity throughout longer sequences remains difficult, particularly for facial features that require fine-grained detail preservation. These limitations are especially pronounced in specialized applications like dancing video generation, where the model must simultaneously handle accurate pose-based motion, temporal consistency, and identity preservation. While models like AnimateAnyone [10] have made progress in this domain, the models are not derived from image generation methods. This highlights the need for more flexible approaches that employ diffusion transformer model.

1.2.6 Summary and Research Gap

Although diffusion models have yielded excellent results in individual domains (image vs. video, style vs. identity), many practical settings require *multi-objective* generation. Existing fusion methods (weight-based, switch-based, noise-based) often assume consistent architectures or sampling schedules. Furthermore, specialized tasks like dancing video generation push current systems to their limits, requiring *temporal stability*, *detailed identity preservation*, and *robust motion* across extensive sequences.

In light of these issues, our work proposes a training-free, **noise-level fusion** mechanism that integrates heterogeneous diffusion models trained with different objectives, architectures, or sampling techniques. This addresses a pressing need for a *flexible, adaptable* pipeline that unifies disparate models' noise predictions without retraining them. Our research fills a vital gap between specialized single-task diffusion models and real-world demands for multi-task, multi-objective generative solutions.

1.3 Contributions and Significance

- **Heterogeneous Model Integration:** In contrast to *ZipLoRA*[11] and *Multi-LoRA Composition*[12], which primarily operate on LoRAs within the same base architecture, our



framework merges *fundamentally different model types*. For example, we can fuse a face-refinement model (Face-Net) with a motion-focused video diffusion model (AnimateAnyone-Net), none of which share perfectly aligned backbones or schedules.

- **Cross-Domain Application:** While *Explorative Inbetweening* [13] and *NVS-Solver* [14] focus on specialized tasks (e.g., novel view synthesis or bounded inbetweening), our *AdaptiveFuse* framework operates in multiple domains: we demonstrate its utility in standard image generation (style vs. identity), in *dancing video* synthesis (merging face-centric and motion-centric models), and in transformer-based video generation (CogVideoX).
- **Adaptive Scheduling Mechanisms:** Beyond simple linear or fixed-fraction mixing, we propose multiple *noise fusion schedulers* to dynamically adjust fusion weights based on the *characteristics of the noise residuals*. This step-varying control is not present in other fusion methods and enables a more delicate balance between style fidelity, identity retention, and motion consistency.

These contributions address the persistent bottlenecks of multi-objective generation, extending the adaptability of diffusion-based pipelines for practical tasks.

1.4 Thesis Outline

The structure of this thesis is as follows:

- **Chapter 1: Introduction.**

Describes the rationale, context, and importance of merging heterogeneous diffusion models. It clarifies the main goals and situates them within today’s research challenges, concluding with a concise summary of our proposed noise-fusion approach.

- **Chapter 2: Literature Review.**

Delves into the theoretical underpinnings of diffusion models and surveys modern applications in image and video generation. It also examines existing model-fusion methods, highlighting their benefits and constraints when managing heterogeneous architectures.



- **Chapter 3: Noise-Fusion in Image-Focused Tasks.**

Demonstrates the noise-level fusion method by combining specialized LoRA modules (e.g., identity LoRA, style LoRA). Implementation details and experimental results show improved output quality without additional training.

- **Chapter 4: Noise-Fusion for Dancing Video Generation.**

Extends the noise-fusion concept to a more demanding application: producing dance videos that require both fluid body movements and accurate subject identity. We discuss how to synchronize different noise schedules and assess the approach via various performance metrics.

- **Chapter 5:** Introduces a novel application of our approach to the CogVideoX framework, demonstrating how multiple specialized LoRAs (camera motion, character identity, and style) can be effectively fused through noise-level fusion. Experimental results show improved text alignment, character consistency, and stylistic quality compared to conventional weight-merging or switching strategies.

- **Chapter 6: Conclusion and Future Work.**

Recaps the main insights gained, outlines the system’s limitations, and suggests directions for expanding noise-fusion into other areas—such as 3D content creation, multi-person interactions, and specialized domain adaptations.

By applying noise-based fusion to diverse tasks, from image stylization to dancing video generation, this work underscores the *originality*, *practicality*, and *scope* of fusing heterogeneous diffusion models in a *training-free*, architecture-agnostic manner. The following chapters include a deeper technical discussion, extensive experimental results, and pathways for further investigation to reinforce and extend the proposed methodology.





Chapter 2. Literature Review

In this chapter, we survey the key developments in diffusion-based generative models. We begin with a discussion of foundational diffusion concepts, reviewing DDPMs, DDIMs, and the well-known Stable Diffusion framework. We then explore state-of-the-art approaches for improving sampling efficiency, integrating various control signals, and personalizing outputs for specific subjects or styles. Next, we discuss existing model-fusion techniques, focusing on the challenges of combining heterogeneous diffusion modules. Finally, we highlight particular applications to dance video generation, illustrating the need for multi-objective solutions that balance motion fidelity and identity preservation.

2.1 Historical Background on Generative Models

Generative modeling has long been a central focus in machine learning, aiming to characterize and sample from complex data distributions. Modern deep generative models have significantly advanced these efforts [15–17, 30–32].

Variational Autoencoders (VAEs). VAEs [15, 30] learn a latent-variable model through an encoder-decoder framework. In this framework, encoder features transform input samples into a latent area, while decoder features reconstruct the samples. While VAEs are relatively stable to train and can handle large datasets, their outputs often exhibit blurriness or reduced sharpness.

Generative Adversarial Networks (GANs). GANs [31] train two distinct networks. One is a generator. The other is a discriminator. These networks operate in an adversarial loop. Variants of GANs [32] have achieved remarkably high-resolution image generation. However, issues such as mode destruction, training fluctuation, and difficulties while controlling outputs remain significant limitations [16].

Flow Models. Normalizing flows [17] convert an uncomplicated distribution (e.g., Gaussian) into a more elaborate one using sequential invertible transformations. Flows enable tractable



likelihood estimation but can become cumbersome for high-resolution tasks due to the requirement of invertible transformations with tractable Jacobians.

Rise of Diffusion Models. Diffusion models [1–3] address many challenges inherent in GANs, VAEs, and flows by structuring generation as a multi-step denoising process. Their iterative nature often yields stable training dynamics and extensive coverage of the target distribution [4, 33]. As detailed in subsequent sections, diffusion models have become a cornerstone in both image and video generation tasks.

2.2 Diffusion Models

Diffusion models form a category of generative approaches that convert random noise into structured data through iterative forward and backward processes [34]. This paradigm has demonstrated exceptional performance in high-quality image, audio, and video synthesis, marking a significant milestone in generative modeling research [1–3]. Beyond these foundational works, theoretical and algorithmic advances have addressed sampling efficiency, scoring functions, and model interpretability [34–36].

2.2.1 Denoising Diffusion Probabilistic Models (DDPM)

DDPMs [33, 37, 38] serve as a foundational framework in this area. Consider a data sample x_0 drawn from an actual sample distribution $q(x_0)$. The *forward diffusion procedure* gradually adds Gaussian noise to produce a sequence $\{x_t\}_{t=1}^T$:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}), \quad (2-1)$$

where β_t defines the noise schedule. Over multiple steps, x_T becomes nearly random noise. One can express x_t directly in terms of x_0 :

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \text{where } \alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{i=1}^t \alpha_i, \text{ and } \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (2-2)$$

As shown in Figure 2.1, this forward process progressively destroys the data structure, turning a clean sample into nearly pure noise. Alternative parameterizations and improved noise schedules have been explored to enhance convergence and sample quality [33, 38].

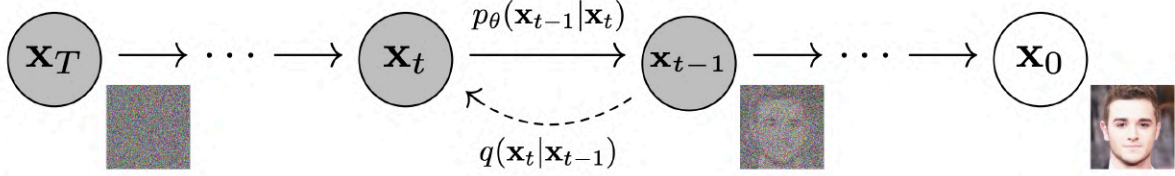


Figure 2.1 Illustration of the forward process in a typical DDPM framework, where the data distribution is gradually corrupted with noise.

The reverse step in this process intends to reconstruct x_0 by training a distribution $p_\theta(x_{t-1}|x_t)$. While the true reverse distribution is intractable, one can train a U-Net to approximate the added noise ϵ at individual steps:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)). \quad (2-3)$$

The value of Σ_θ is generally fixed or learned to simplify training in practice. The model is learned through the minimization of the below loss function, which measures the difference among the approximated noise $\epsilon_\theta(x_t, t)$ and the actual noise ϵ :

$$\mathcal{L} = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2]. \quad (2-4)$$

Various loss reweighting and architectural refinements have also been studied to improve training stability and model expressiveness [39, 40].

2.2.2 Training and Sampling

Training involves sampling (x_0, t, ϵ) , computing x_t , and updating θ to minimize the noise prediction error. Once trained, sampling (generation) starts from $x_T \sim \mathcal{N}(0, \mathbf{I})$ and applies the reverse diffusion process until x_0 is obtained:

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z, \quad z \sim \mathcal{N}(0, \mathbf{I}),$$

Improvements in sampling procedures [36, 41] strive to decrease the denoising iterations as well as preserve high output quality.

2.2.3 Denoising Diffusion Implicit Models (DDIM)

DDIMs [37] introduce an adjustable parameter η , enabling a deterministic or more flexible reverse process. When $\eta = 0$, the sampling transforms into a deterministic mapping, accelerating sampling and enabling easier manipulation such as DDIM inversion. Exploiting DDIM inversion, techniques like Prompt-to-Prompt editing [42, 43] enable sophisticated image editing directly in the denoising procedure. The DDIM reverse steps are defined as:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_t, \quad (2-5)$$

with adjusted parameters ensuring a consistent reverse mapping. This allows faster and more stable sampling and provides a pathway to advanced editing techniques by navigating in the latent (or intermediate) spaces of the model.

2.3 Stable Diffusion Model

While DDPMs and DDIMs produce impressive results in pixel space, they can be computationally expensive, especially at high resolutions. The Stable Diffusion model [5] addresses this challenge by moving the diffusion steps into a latent space, thereby reducing computational overhead without sacrificing output fidelity. Other latent diffusion frameworks and improvements are discussed in [44, 45].

Variational Autoencoder (VAE)

A key component of Stable Diffusion is a VAE (E, D) . The encoder E compresses an image $x \in \mathbb{R}^{H \times W \times 3}$ into a latent code $z = E(x)$. On the other hand, the decoder D reconstructs an image $\hat{x} = D(z)$. This latent representation is generally much smaller than the original image space, making subsequent diffusion steps more efficient.

Latent Diffusion Process

Stable Diffusion generates images by starting from random noise in latent space and iteratively refining it into a coherent representation z_0 over T timesteps. In more detail, the forward process adds noise to an initial latent code z_0 :

$$z_t = \sqrt{\alpha_t} z_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i,$$

where α_t is a schedule hyperparameter and $\epsilon \sim \mathcal{N}(0, I)$. During training, the network ϵ_θ (often a U-Net) is taught to anticipate the noise added at individual step t . The training objective:

$$\mathcal{L} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0, I), t} \left[\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2 \right], \quad (2-6)$$

minimizes the discrepancy between predicted and actual noise, where c is a conditioning vector, like a CLIP text encoder [46].

Inference (Reverse Diffusion): In the inference phase, one starts with a random latent z_T drawn from a Gaussian distribution and applies the learned reverse transitions at each timestep. After T iterations, the final latent code z_0 is obtained, which is passed to the VAE decoder D , yielding a high-resolution output $\hat{x} = D(z_0)$. This latent-space denoising framework brings significant efficiency gains over pixel-space diffusion, particularly at higher resolutions.

U-Net with Cross-Attention for Conditional Guidance

The core denoising step in Stable Diffusion uses a U-Net architecture, selected for its efficacy in multi-scale feature extraction and reconstruction. As shown in Figure 2.2, the U-Net's symmetrical design with skip connections helps preserve both local details and global context, ensuring fine-grained and coherent generation. Architectural and sampling refinements for U-Net-based diffusion models appear in [35, 36].

To enable text-conditioning, Stable Diffusion integrates a pre-trained text encoder, like CLIP, and incorporates its embeddings into the U-Net via cross-attention. Let y be a text prompt and $\tau(y)$ its embedding. During denoising, cross-attention aligns image features $\phi_i(z_t)$ with

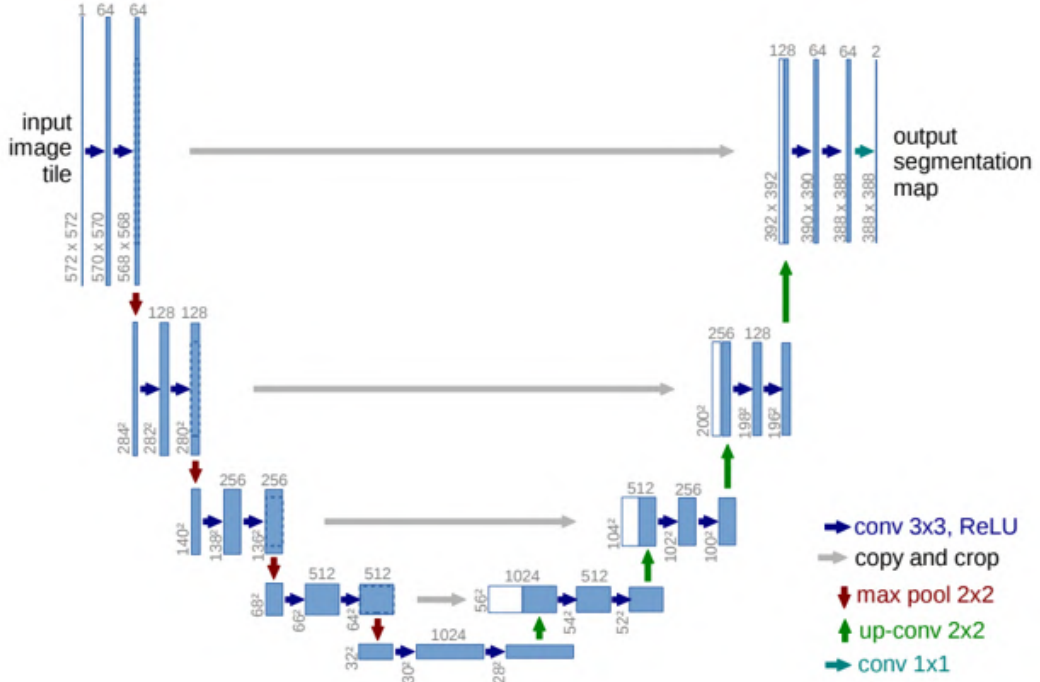


Figure 2.2 A U-Net architecture used by Stable Diffusion. Skip links bridge encoder and decoder, helping retain spatial detail and improving reconstruction quality.

textual representations $\tau(y)$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V, \quad (2-7)$$

where

$$Q = W_Q \phi_i(z_t), \quad K = W_K \tau(y), \quad V = W_V \tau(y),$$

and W_Q, W_K, W_V are trainable variables. This mechanism infuses semantic guidance into the diffusion steps, enabling flexible text-driven or multimodal generation. Refinements to prompt conditioning and multimodal guidance have been explored in [47, 48].

In summary, Stable Diffusion leverages an autoencoder to compress images into a latent code, mitigating the computational burden of pixel-space diffusion. A U-Net operating on these latent representations performs iterative denoising, guided by cross-attention for flexible conditioning. This paradigm stands as a significant advancement over earlier approaches, enabling highly efficient and controllable image synthesis. It provides a cornerstone for subsequent developments that extend diffusion models to more specialized tasks, such as personalized stylization, inpainting, and high-resolution video generation.

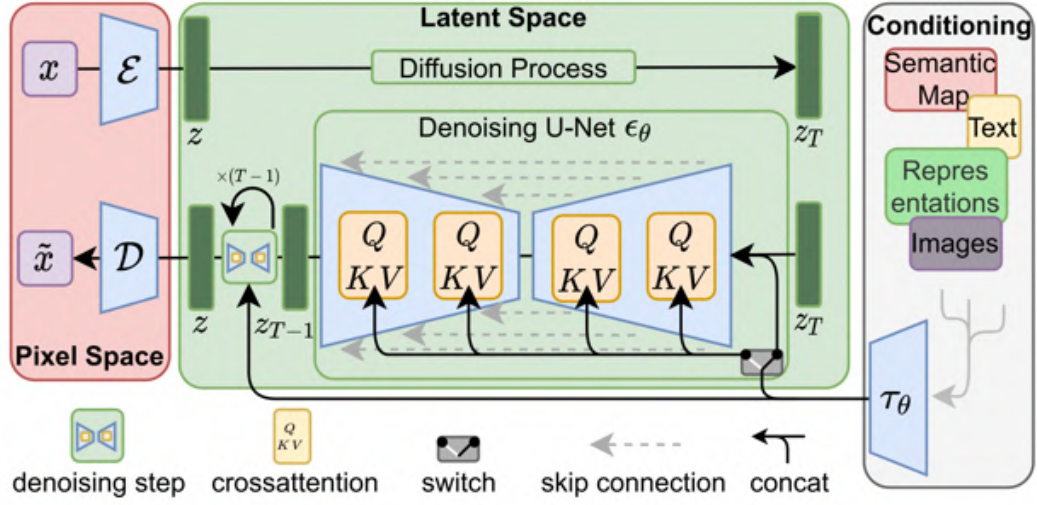


Figure 2.3 Pipeline for Stable Diffusion. First, a text prompt is transformed into an embedding by a text encoder. Next, diffusion happens in the latent space under cross-attention from text-image interactions. Finally, the VAE decoder reconstructs the high-resolution image from the refined latent code.

2.4 Diffusion Models for Image Generation

Diffusion-based strategies have led to cutting-edge advancements in image generation, surpassing conventional GAN-based techniques in variety and fidelity. The versatility of diffusion models has led to their adoption in a wide spectrum of tasks, like image generation [1–5, 18, 33, 49], image super-resolution [5, 33, 40, 50–52], image inpainting [1, 3, 5, 18, 52, 53], image translation [54], and image editing [55–57]. Conditioning mechanisms further boost performance on complex datasets [54, 58].

Low-Rank Adaptation (LoRA): LoRA [8] is a low-rank fine-tuning method originally used in language models but later adapted for diffusion. It introduces small, rank-constrained weight matrices into the existing layers, allowing the model to pick up new styles or identities with minimal additional parameters. LoRA drastically reduces training time and memory usage compared to retraining the entire network.

ControlNet: ControlNet [20] adds a parallel trainable copy of Stable Diffusion’s encoder-decoder blocks to process auxiliary inputs, such as edges, segmentation maps, pose heatmaps. As shown in Figure 2.4, these copied modules run alongside the original U-Net and are merged via *zero convolution* layers, which initialize to zero weights so that the base model remains

unchanged at first. During training, the model learns how much of the ControlNet branch's features to incorporate for fine-grained control. This design yields a flexible framework for integrating diverse spatial conditions without overwriting the powerful priors in the pre-trained Stable Diffusion weights, and has inspired further work on multimodal guidance and complex conditioning [59, 60].

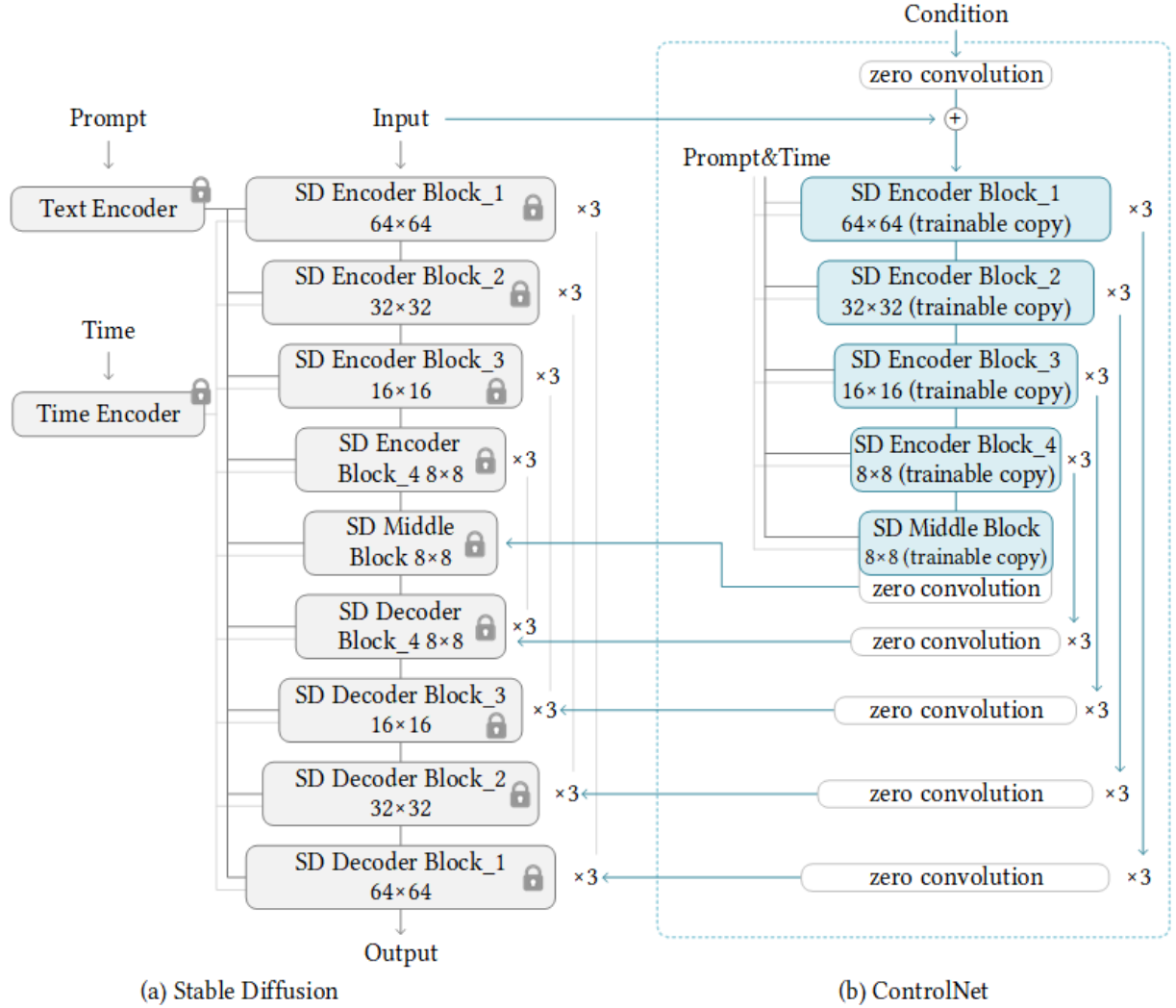


Figure 2.4 Overview of ControlNet. (a) Stable Diffusion layers remain frozen (gray locks). (b) A parallel trainable branch, equipped with zero-convolution blocks, injects conditional features (edges, poses, etc.) into the main U-Net.

IP-Adapter: IP-Adapter [21] fuses image and text prompts for personalized, multi-conditioned generation. As shown in Figure 2.5, a frozen image encoder extracts reference-image features, which are then projected into the diffusion U-Net. Meanwhile, a frozen text encoder processes text prompts. By *decoupling cross-attention*, IP-Adapter allows image-derived cues, such as

style or subject identity, and text guidance (semantic constraints) to coexist without conflict. Similar methods [48] refine prompt conditioning, but IP-Adapter stands out for seamlessly integrating image-based cues into the diffusion process for more flexible personalization.

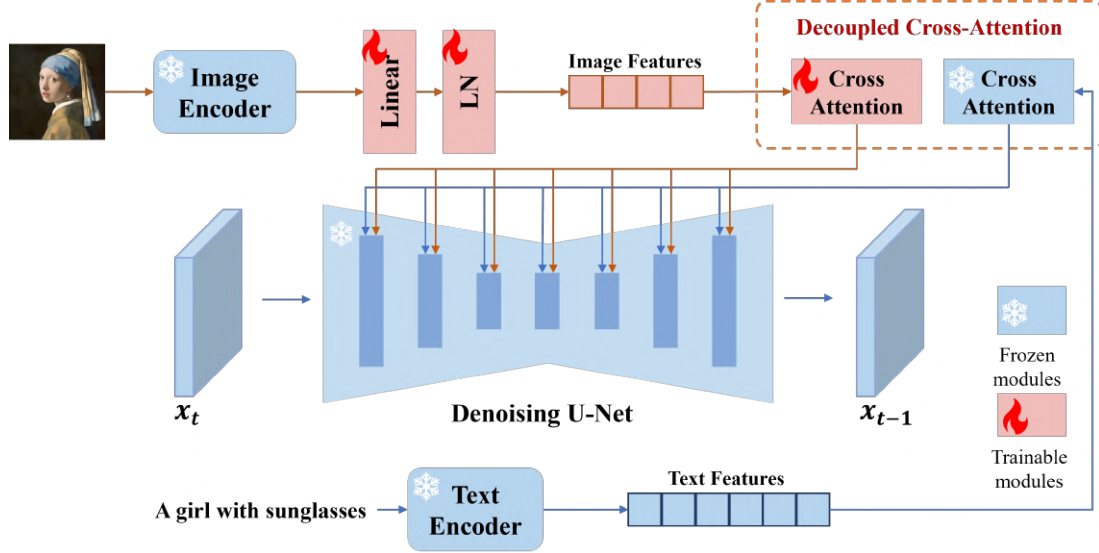


Figure 2.5 Overview of IP-Adapter. Frozen modules (blue) encode images and text, while trainable blocks (red) project image features into the U-Net, enabling decoupled cross-attention.

IP-Adapter FaceID: Building on the IP-Adapter design, IP-Adapter FaceID [21] further refines facial attributes, aligning generated faces with reference identities. Similar image-to-image translation techniques [54, 58] also enhance personalization.

Style Transfer and Identity Preservation: Techniques like StyleDrop [9] and Instant ID [22] focus on transferring specific styles or preserving identities. Additional personalization approaches, such as DreamBooth [61] and Textual Inversion [62], learn new concepts from a handful of images, thereby expanding the model’s versatility.

2.5 Diffusion Models for Video Generation

Building upon LDMs for image generation [4, 5, 19], recent work has explored methods to animate text-to-image (T2I) pipelines for temporal tasks.

AnimateDiff. *AnimateDiff* [7] (Figure 2.7) inserts a *plug-and-play motion module* into a personalized T2I U-Net without re-training the original weights. This module, trained on real-world videos, injects motion priors that yield coherent animations aligned with the model’s specialized appearance or content knowledge.

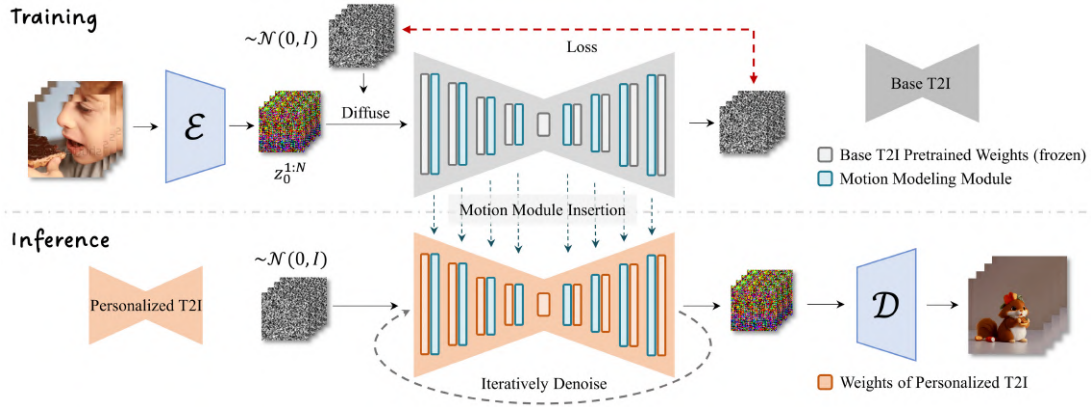


Figure 2.6 AnimateDiff inserts a single motion module (highlighted) into a pre-trained T2I model, enabling temporal coherence without overwriting existing appearance priors.

Human-Centric Video Generation. Building on the motion module architecture, recent works focus particularly on human-centric video generation, such as dance or pose-driven animation [10, 23–25, 63, 64], where fine-grained details like facial expressions and body shape must remain consistent throughout the video. Despite the progress, preserving detailed facial features remains an open challenge; subtle deviations in early frames often compound over time, leading to flickering or drift. Ongoing efforts to address this issue include integrating stronger identity constraints [65] or leveraging disentangled pose-and-appearance representations [66, 67] to stabilize the generation process.

Editing and Adaptation. Other methods, such as Tune-A-Video [68] and Video-P2P [69], extend diffusion-based editing to the video domain, fine-tuning pre-trained T2I or latent-diffusion models on short video clips for consistency across frames. FateZero [70] aligns generative in-

variances to reduce flicker, while MagicAnimate [23] uses specialized condition encoders to better preserve subject appearance. Although promising, these methods often require specialized training or large-scale adaptation to handle diverse motions and complex scenes.

Overall, diffusion-based video generation has shown promise for complex, multi-frame tasks, yet further research is needed to handle longer sequences and ensure robust identity preservation, especially in demanding scenarios like dance and motion-intensive actions.

2.6 Diffusion Transformer model

Video generation represents one of the most challenging and computationally intensive applications of generative models. Several state-of-the-art systems employ transformer-based diffusion models to tackle this challenge [30, 71–73]. OpenAI’s Sora [71] uses a space-time diffusion transformer to generate remarkably realistic videos from text prompts, with consistent motion and coherent global structure. Similarly, VideoPoet [72] and CogVideoX [29] utilize transformer architectures to model both spatial and temporal relationships simultaneously.

The key advantage of DiTs for video generation lies in their ability to model long-range dependencies across both the spatial and temporal dimensions. This capability is crucial for maintaining consistency in camera movements, character appearances, and scene dynamics throughout extended sequences.

2.6.1 Multi-Modal Conditioning

DiT architectures readily accommodate multiple conditioning signals through cross-attention mechanisms or adaptive normalization layers. This flexibility enables precise control over generation through various guiding signals. For example, DALL-E 3 [19] employs cross-attention mechanisms to incorporate text embeddings, while models like CogVideoX [29] use ”Expert Adaptive LayerNorm” to facilitate fusion between text and visual modalities. The transformer architecture is particularly well-suited for handling heterogeneous conditioning inputs, as it can dynamically attend to relevant parts of each condition signal based on the generation context. This capability enables more nuanced control over the generative process compared to earlier

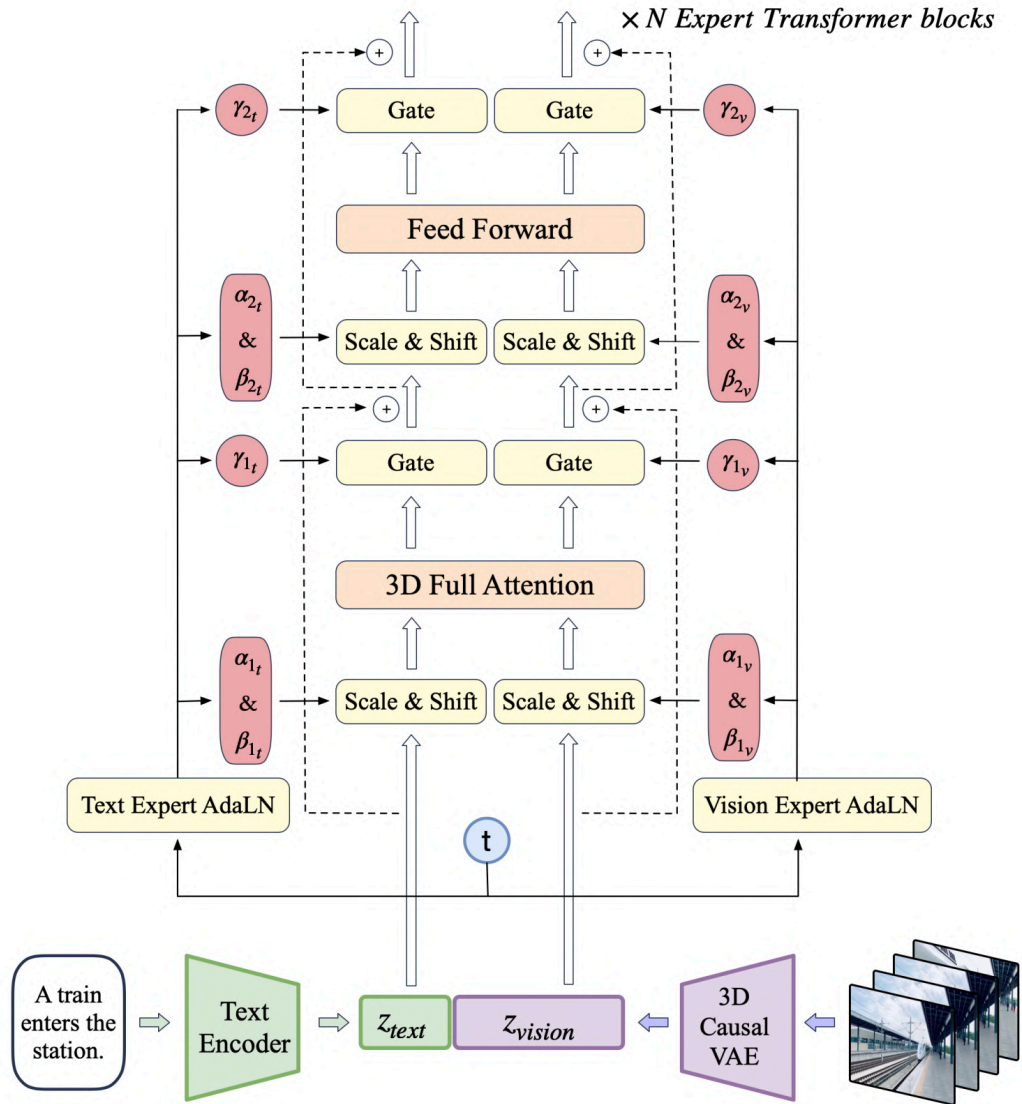


Figure 2.7 CogVideoX has 3D Causal Variational Autoencoder (VAE) and Expert Transformer, enabling longer sequence length and deeper text embeddings.

diffusion models.

2.6.2 Personalization and Adaptation

Transformer-based diffusion models have demonstrated remarkable adaptability through parameter-efficient fine-tuning techniques [74]. Methods like LoRA [8] are frequently applied to DiTs to create personalized generators that preserve a subject’s identity or adopt specific artistic styles. The structured nature of transformer architectures makes them amenable to targeted adaptations. Recent work has explored more advanced personalization techniques specifically tailored for DiTs, leveraging the inherent modularity of attention mechanisms to introduce specialized knowledge without compromising the model’s general capabilities [73, 75–80].

The application of DiTs continues to expand rapidly, with researchers exploring increasingly ambitious generative tasks that benefit from the model’s strong representational capacity and flexibility. As computational resources and architectural innovations advance, DiT-based approaches are likely to remain at the forefront of generative AI research and applications.

2.7 Model Fusion Approaches

Model fusion strategies aim to combine the strengths of multiple pre-trained models (or specialized modules) to achieve various objectives simultaneously. Broadly speaking, fusion methods merge or integrate parameters, modules, or noise residuals from distinct diffusion models (or specialized LoRA modules) into a single generative pipeline. In this section, we provide an extended discussion of three major fusion paradigms, such as weight-based, switch-based, and noise-based, highlighting recent advances such as *ZipLoRA* [11], *Multi-LoRA Composition* [12], *NVS-Solver* [14], and *Explorative Inbetweening of Time and Space* [13] to illustrate the breadth of existing solutions and the remaining challenges.

2.7.1 Weight-Based Fusion

Weight-based fusion involves directly merging model parameters, such as the main diffusion U-Net weights or LoRA matrices, typically by combining and re-scaling their parameters.

This class of approaches is the most straightforward for unifying multiple specialized models.

General Formulation. Let θ_1 and θ_2 denote the parameter sets, like LoRA weights, of two pre-trained diffusion models, each specializing in different tasks or styles. A naive linear weight-based merge computes a fused parameter set:

$$\theta_{\text{fused}} = \alpha \theta_1 + (1 - \alpha) \theta_2, \quad (2-8)$$

where $\alpha \in [0, 1]$ is a user-defined scalar.

ZipLoRA [11]. ZipLoRA provides an effective approach to fuse multiple LoRA modules trained on different subjects or styles. It identifies and reduces column-wise overlaps in the LoRA matrices by learning merger coefficients that mitigate destructive interference among highly correlated columns:

$$\Delta W_{\text{Zip}} = \Delta W_1 \mathbf{Q}_1 + \Delta W_2 \mathbf{Q}_2 + \dots \quad (2-9)$$

where each ΔW_i is a LoRA update matrix, and $\mathbf{Q}_i$ is an adjustment, like learned re-scaling, to handle overlaps. Despite its efficacy, ZipLoRA assumes all LoRAs share identical base architectures and training schedules, thus limiting its applicability if the underlying diffusion models differ in design or hyperparameters.

Observations. Weight-based fusions provide a direct means of weights merging. They have minimal overhead at inference, since only one model is produced, but often exhibit limited flexibility if the base models have distinct architectures.

2.7.2 Switch-Based Fusion

Switch-based fusion dynamically selects among multiple specialized diffusion branches at different timesteps during the reverse diffusion process. Instead of merging parameters, the approach orchestrates specialized skill modules by swapping or activating one at a time, typically to blend multiple styles or subjects in a single generation.

General Formulation. Let $\{\theta_1, \theta_2, \dots, \theta_k\}$ denote the parameter sets for k specialized modules (LoRAs). During the denoising step t , one might only activate module i with probability or schedule $s_t(i)$. The forward pass in the diffusion model with switch-based fusion can be written as:

$$\Phi(\mathbf{x}_t, t) = \sum_{i=1}^k s_t(i) \Phi(\mathbf{x}_t | \theta_i), \quad (2-10)$$

where $s_t(i)$ are typically one-hot over i , implying only one active module at each step.

LoRA Composite [12]. Proposed in the context of *Multi-LoRA Composition*, LoRA Composite activates distinct LoRA modules at selected denoising steps. For instance,

$$s_t(i) = \begin{cases} 1, & \text{if } (t \bmod \tau) = i, \\ 0, & \text{otherwise.} \end{cases}$$

This ensures that each LoRA is applied during specific intervals of reverse diffusion sampling. While effective, this method can only activate one of the LoRA modules, resulting in unstable generation results, especially when working with fine-grained generation.

2.7.3 Noise-Based Fusion

Noise-based fusion manipulates the *noise residuals* predicted by multiple models or specialized modules, rather than merging or selecting their parameters. Typically, each model runs in parallel at each denoising step, and their noise predictions are *combined* (averaged or weighted).

General Formulation. Let $\epsilon_{\theta_1}(\mathbf{x}_t, t)$ and $\epsilon_{\theta_2}(\mathbf{x}_t, t)$ be the predicted noise from two distinct diffusion modules. A noise-based fusion approach might produce:

$$\hat{\epsilon}_t = \alpha \epsilon_{\theta_1}(\mathbf{x}_t, t) + (1 - \alpha) \epsilon_{\theta_2}(\mathbf{x}_t, t), \quad (2-11)$$

which is then used to update $\mathbf{x}_{t-1}$. This procedure can be extended to more than two modules by summing over multiple noise estimates.

NVS-Solver [14]. A zero-shot approach that fuses the noise predictions from a base *video diffusion model* with *3D geometry constraints*, ensuring multi-view consistency for novel view synthesis. Although not explicitly framed as fusing separate modules, it effectively merges different noise signals to adapt the model at inference:

$$\epsilon_{\text{fused}}(\mathbf{x}_t, t) = \epsilon_{\theta_{\text{vid}}}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \mathcal{L}_{\text{geom}}(\mathbf{x}_t), \quad (2-12)$$

where $\mathcal{L}_{\text{geom}}$ is a geometry-based prior. This perspective underscores the flexibility of noise-based strategies for combining multiple constraints.

Explorative Inbetweening of Time and Space [13]. The method of *Time Reversal Fusion* is employed to generate two parallel denoising trajectories: one propagating forward in time, given a starting frame, and the other propagating backward, given an ending frame. By either averaging or optimizing the partial noise estimates, the following equation defines the resulting fusion:

$$\epsilon_t^* = \alpha_t \epsilon_{\text{forward}}(\mathbf{x}_t, t) + (1 - \alpha_t) \epsilon_{\text{backward}}(\mathbf{x}_t, t), \quad (2-13)$$

it merges the two constraints into one coherent trajectory that begins in the start frame and ends in the target end frame. This exemplifies how noise-based fusion can incorporate multiple distinct boundary conditions (start vs. end frames) without re-tuning the underlying diffusion model.

Key Properties. Compared to weight- or switch-based methods, noise-level fusion (Eq. 2-11) avoids direct parameter conflicts. Instead, it manipulates each model’s final noise estimate, thus often preserving each model’s specialized knowledge more robustly. Nevertheless, large discrepancies in the predicted noise from different modules can introduce sampling artifacts or oscillations, making the design of the fusion weight α_t crucial. In addition, these methods only address novel view synthesis.

2.7.4 Challenges in Heterogeneous Fusion

Although weight-based, switch-based, and noise-based fusion each offer unique advantages, several open challenges arise when practitioners aim to merge *heterogeneous* diffusion

models or specialized LoRA modules trained on diverse tasks or data distributions:

- **Architecture Mismatch.** Merging modules at the weight level (weight-based fusion) often presupposes a consistent U-Net architecture or uniform training schedule. Heterogeneous models, such as a base diffusion model for stylization plus a specialized video diffusion model for temporal consistency, pose architectural discrepancies that hamper direct merges.
- **Divergent Noise/Time Schedules.** When one model is trained on a different timestep schedule or with distinct noise distributions, naively averaging noise residuals can create sampling artifacts. Switch-based or weight-based methods also struggle unless they unify the sampling procedure across models.
- **Subject or Style Interference.** Even if the base architecture matches, simply adding LoRA updates can cause destructive interference when two modules attempt to manipulate the same parameters or latent features in conflicting ways. More advanced methods, like ZipLoRA, attempt to mitigate this, but multi-LoRA composition with many specialized modules is still an open challenge [12].

Overall, fusing multiple diffusion models (or LoRA modules) remains a rapidly evolving frontier in controllable generative modeling. Methods such as *ZipLoRA* [11], *Multi-LoRA Composition* [12], *NVS-Solver* [14], and *Explorative Inbetweening of Time and Space* [13] showcase promising strategies spanning all three major paradigms. However, there are challenges to employing the fusion methods, designing more robust weighting mechanisms, and extending them to heterogeneous backbones with mismatched architectures, noise schedules, or domain objectives.

2.8 Applications in Dance Video Generation

Dance video generation, wherein a human dancer’s pose, motion, and identity are synthesized or transferred to produce novel, coherent dancing sequences, has drawn notable attention

in computer vision and multimedia research. The task has practical applications in the entertainment industry, social media content creation, and digital choreography. This section introduces an extended summary of current advances in dance video generation, focusing on four representative methods, such as *DisCo* [24], *MagicPose* [25], *MagicAnimate* [23], and *AnimateAnyone* [10], to illustrate both shared fundamentals and unique design choices. We then situate our own approach in context, discussing how it connects with existing works and highlighting the novel aspects.

2.8.1 Key Challenges in Dance Video Generation

Generating realistic dance videos requires addressing multiple interlinked challenges:

1. **Motion Fidelity and Diversity:** The model should capture a wide range of poses and ensure realistic, temporally consistent movements.
2. **Identity Preservation:** Facial features, clothing details, and body proportions of a reference identity must be retained across a sequence of frames.
3. **Temporal Consistency:** Minimizing flicker or jitter artifacts is crucial for smooth motion.
4. **Generalization to Unseen Domains:** Real-world social media dance scenarios involve diverse backgrounds, camera viewpoints, and subject appearances not seen in training data.
5. **Compositionality and Modularity:** Real applications often demand free combination of backgrounds, motion signals, and subject appearances without exhaustive re-training. This requires carefully disentangled representations for pose and appearance.

2.8.2 Recent Methods

DisCo: Disentangled Control for Realistic Human Dance Generation [24]. DisCo adopts a strategy of *disentangled control*, ensuring backgrounds, subjects, and skeleton-based motion can each be managed independently. It leverages a multi-stage training scheme that integrates



ControlNet [20] with large-scale fine-tuning on human attributes. This approach fosters compositional synthesis, such as combining arbitrary backgrounds with novel subjects and poses. While DisCo excels at per-frame realism and identity retention, it employs relatively simple single-subject scenarios and primarily focuses on single-person dance.

MagicPose: Realistic Human Poses and Facial Expressions Retargeting [25]. Targeting accurate *pose* and *facial expression* transfer, MagicPose incorporates two parallel condition streams: skeleton heatmaps for body poses and facial landmark encodings for expression. A separate *appearance control block* blends reference-image features into the diffusion process, improving identity preservation (including subtle face details). This technique excels at generating high-fidelity close-up views where facial attributes matter significantly, though less emphasis is placed on backgrounds or multi-object scenarios.

MagicAnimate: Temporally Consistent Human Image Animation [23]. MagicAnimate centers on ensuring *temporal stability* across frames. A video-based diffusion backbone is equipped with explicit temporal attention modules, preventing frame-to-frame flicker. Additionally, a specialized *appearance encoder* refines identity details and a video fusion strategy mitigates abrupt transitions. MagicAnimate demonstrates improved smoothness for long dance sequences but can face challenges when posed with significantly different or unseen human identities.

AnimateAnyone: Consistent and Controllable Image-to-Video Synthesis [10]. AnimateAnyone extends pose-based controllability to a broader character animation setting. By unifying appearance encoding, motion retargeting, and spatiotemporal consistency in one pipeline, it aims at a highly reusable framework for various types of subjects (including stylized characters). Its multi-stage approach emphasizes identity consistency across transformations and can adapt to different motion signals beyond dance poses. Fine-grained details such as complex hairstyles and hand gestures remain partially open for further refinement.

2.8.3 Comparisons and Distinctions among Recent Works

While these methods share a foundation in diffusion-based generators, they differ in how they encode reference identity, integrate pose constraints, and address video-level consistency:

- **Identity Encoding:** DisCo and AnimateAnyone employ CLIP-based image embeddings for the reference subject, while MagicPose and MagicAnimate each introduce a specialized appearance module.
- **Temporal Modeling:** MagicAnimate and AnimateAnyone explicitly model multi-frame temporal consistency, while DisCo and MagicPose focus on single-image frameworks with implicit or partial spatiotemporal constraints.
- **Background Integration:** DisCo treats background composition independently, MagicPose, MagicAnimate and AnimateAnyone rely on holistic diffusion backbones for background alignment.

2.8.4 Potential for the AdaptiveFuse method

Even with robust motion synthesis, subtle facial features often degrade over multiple frames, leading to flicker or identity drift [10, 65]. Dedicated face modules, like IP-Adapter FaceID [21], or specialized face LoRAs can help maintain fidelity, but typically demand additional training or complex post-processing steps.

As these methods illustrate, dance video generation can benefit significantly from combining specialized modules for pose and motion, style or identity preservation, and face refinement. However, a single diffusion model trained end-to-end on all objectives often struggles to excel at every requirement without vast data or extensive fine-tuning. *Model fusion* strategies offer a potential solution, allowing practitioners to leverage multiple pre-trained modules that excel in different aspects, such as motion realism, facial detail, or stylization. Yet, the mismatch in architectures or noise schedules remains a substantial challenge.

2.9 Application in Diffusion Transformer model

Diffusion Transformer (DiT) models, particularly CogVideoX, present unique opportunities and challenges when integrating multiple specialized capabilities. While CogVideoX demonstrates impressive text-to-video generation capabilities, several limitations emerge when attempting to simultaneously handle multiple objectives like camera motion, stylization, and character identity preservation.

2.9.1 Multi-Objective Challenges in CogVideoX

CogVideoX, with its expert transformer architecture and 3D causal VAE, provides a powerful foundation for video generation. However, when adapting CogVideoX for specialized tasks requiring multiple capabilities, several gaps become apparent:

- **LoRA Integration Limitations:** While CogVideoX supports LoRA fine-tuning, naively loading multiple LoRAs (e.g., for camera motion, style, and character identity) into a single transformer often results in parameter conflicts and compromised generation quality.
- **Identity-Style Trade-offs:** When attempting to simultaneously preserve character identity (e.g., specific facial features) while applying strong stylistic elements (e.g., black and white, anime-style), single-pipeline approaches frequently sacrifice one aspect for another.
- **Dynamic Control Requirements:** Different aspects of generation may benefit from varying influence at different denoising steps—for instance, establishing motion and composition early while refining stylistic details later. Standard CogVideoX lacks mechanisms for such fine-grained control.

2.9.2 Motivation for Multi-LoRA Fusion

These limitations motivate the development of novel fusion strategies specifically tailored for CogVideoX and other DiT architectures. Unlike convolutional models, transformer-based architectures present distinct challenges and opportunities for model fusion:

- The self-attention mechanisms central to transformers create complex, global parameter interdependencies that make naive weight merging particularly problematic.
- The large parameter count of transformer models (e.g., CogVideoX-5B) makes full fine-tuning for each specialized task impractical, necessitating more efficient adaptation and fusion approaches.

Given these considerations, there is a clear need for fusion strategies that can integrate multiple specialized capabilities in CogVideoX without destructive interference or extensive retraining. By addressing these challenges through adaptive noise-level fusion rather than parameter-space merging, we can potentially harness the full power of specialized LoRAs while preserving the strengths of the underlying transformer architecture.

As demonstrated in dance video generation applications, even state-of-the-art models struggle to simultaneously maintain motion fidelity, identity preservation, and stylistic consistency. This gap underscores the potential impact of successful multi-LoRA fusion techniques for CogVideoX and similar transformer-based video diffusion models.

2.10 Summary of Literature Review

Diffusion models have substantially advanced both image and video generation, offering stable training and high-fidelity results. Techniques like DDPMs and Stable Diffusion streamline large-scale synthesis, while modules such as ControlNet, IP-Adapter, and LoRA provide finer-grained control for style, identity, and spatial guidance. Video-oriented strategies, including AnimateDiff and other latent video diffusion models, extend these capabilities to multi-frame settings but introduce new challenges, notably in preserving identity and motion coherence.

Various *model fusion* techniques (weight-based, switch-based, noise-based) enable multi-objective generation but often assume homogeneous architectures or noise schedules, limiting their application to heterogeneous tasks. Meanwhile, tasks such as dance video generation exemplify the need for advanced multi-condition pipelines that simultaneously handle pose fidelity, style personalization, and face-centric consistency.



These observations underscore the need for a more flexible, training-free, and *model-agnostic noise-fusion* framework capable of merging heterogeneous diffusion modules. The proposed *AdaptiveFuse* approach, presented in subsequent chapters, aims to address these gaps by reconciling multiple specialized models, such as face-focused and motion-focused diffusion, to produce high-quality, identity-preserving dance videos without extensive re-training or strict architectural alignment.





Chapter 3. Character LoRA and Style LoRA Fusion

Before addressing the dance video generation, this chapter introduces a novel noise-based fusion strategy, referred to as **AdaptiveFuse**, that integrates multiple LoRAs, specifically Character LoRA and Style LoRA, within a single image diffusion model. By operating at the noise residual level, rather than merging model weights directly, AdaptiveFuse offers a flexible and adaptive framework to maintain both a character’s unique identity and a chosen artistic style. The following sections detail the motivation, methodological approach, experimental setup and result. These findings highlight how noise-based fusion can surpass traditional methods in achieving both identity consistency and stylistic richness in image generation.

3.1 Motivation

Real-world image-generation applications often demand maintaining *multiple* specialized capabilities in a single model. For instance, users may require a diffusion model to depict a specific character or subject consistently (e.g., a well-known celebrity), while also applying a distinctive artistic style such as line art or oil painting. LoRA [8] provides a powerful way to fine-tune large diffusion backbones on niche tasks, including identity preservation or stylistic rendering, without the overhead of re-training the entire model. However, simply merging multiple LoRA modules via direct parameter interpolation often leads to compromised results, where either the style is weakened or the unique identity is blurred [11].

Goal. Our aim is to seamlessly combine *Character LoRA* and *Style LoRA* so that both the subject’s identity and the target art style remain unaltered. In other words, we seek to generate images that faithfully capture an individual’s facial features, expression, and attire, while simultaneously rendering these characteristics in a specific stylistic domain.

3.2 Methodology

3.2.1 AdaptiveFuse Framework

AdaptiveFuse introduces an innovative approach for combining multiple LoRAs via noise residual manipulation during the denoising process. In contrast to traditional approaches that directly merge LoRA weights, AdaptiveFuse operates at the noise level, enabling a model-agnostic and more dynamically adaptable synthesis.

As illustrated in Figure 3.1 (a), our framework features two LoRA-augmented diffusion U-Nets: the **Character LoRA-Net** (top) and the **Style LoRA-Net** (bottom). Both U-Nets receive pose inputs from ControlNet (via OpenPose keypoints) to ensure that the synthesized image respects the intended body posture. Additionally, text embeddings from a text CLIP encoder guide each network’s cross-attention layers, informing the overall visual theme or semantic content.

The Character LoRA-Net incorporates a LoRA module designed to preserve the subject’s identity and key features, injecting low-rank parameter updates into the spatial and cross-attention. In parallel, the Style LoRA-Net applies a LoRA module optimized for rendering a particular artistic style (e.g., oil painting or sketch). Both of these specialized U-Nets produce their own noise residuals at each denoising step.

These residuals are then passed to the **Fusion Module**, which adaptively combines them according to the selected fusion strategy, as illustrated in Figure 3.1 (b). The fused latent is iteratively denoised until the final timestep, after which a decoder converts the latent into the final synthesized image. By segregating identity-related adaptations (top U-Net) from style-specific ones (bottom U-Net), our approach allows for precise control over both subject fidelity and artistic effect, while remaining compatible with the pose constraints specified by ControlNet.

3.2.2 Noise-based Fusion Strategies

AdaptiveFuse adopts a noise-level fusion philosophy inspired by approaches such as NVS-Solver [14] and Time Reversal Fusion [13], which highlight that noise-space manipulations can

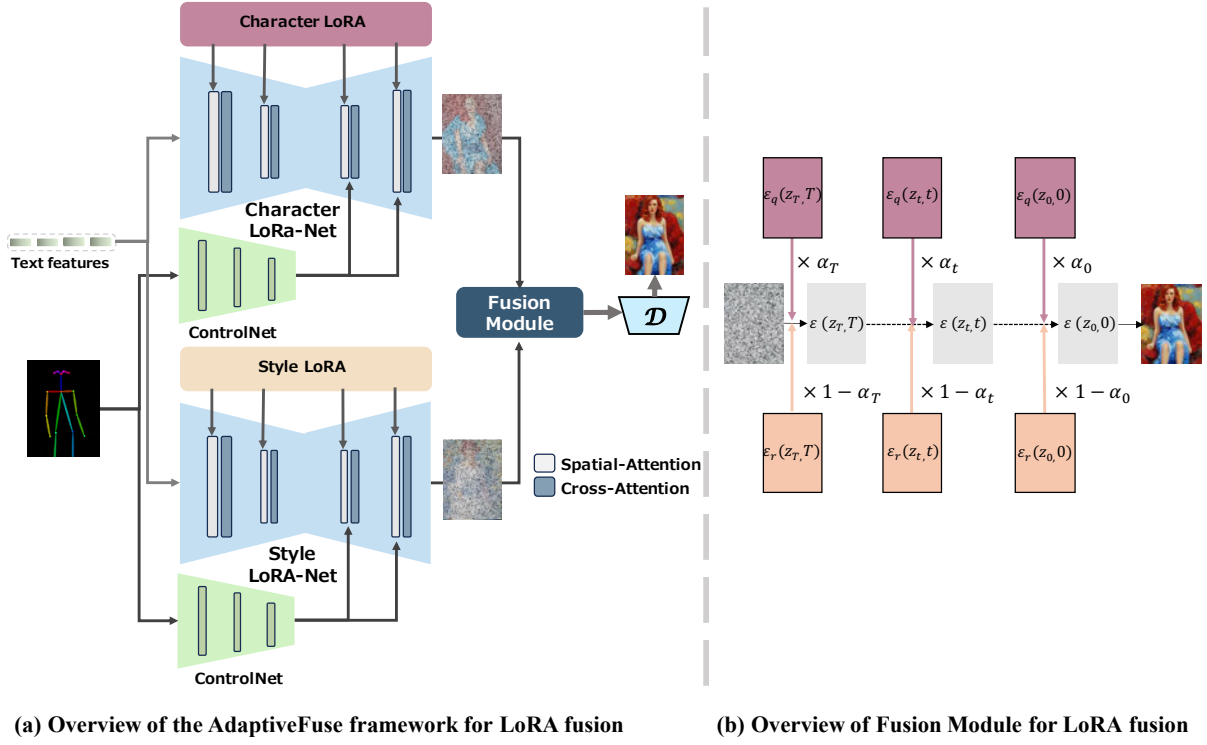


Figure 3.1 Overview of the AdaptiveFuse framework for LoRA fusion. (a) Architecture of AdaptiveFuse. For Character LoRA-Net, Character LoRA is applied, and ControlNet OpenPose encodes the pose sequence. Meanwhile, Style LoRA-Net is employed for rendering specific styles. Both U-Nets compute noise residuals, which are integrated by the Fusion Module. The final denoised image is decoded by the VAE into the generated output. (b) Overview of the Fusion Module. The noise residuals from the Character LoRA-Net and Style LoRA-Net are combined dynamically based on the fusion parameter (α). This process balances the contributions of each LoRA module to the final denoised output.

unite multiple diffusion models with minimal conflict. Rather than blending network weights, AdaptiveFuse dynamically combines the noise residuals yielded by each LoRA-augmented U-Net.

At each timestep t , the two models predict noise residuals $\epsilon_q(\mathbf{z}_t, t)$ from Character LoRA-Net, and $\epsilon_r(\mathbf{z}_t, t)$ from Style LoRA-Net. AdaptiveFuse produces a fused latent:

$$\epsilon(\mathbf{z}_{t-1}, t-1) = \alpha_t \times \epsilon_q(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_r(\mathbf{z}_t, t), \quad (3-1)$$

where α_t is an adaptively chosen or user-specified fusion parameter.

3.2.3 Fusion Strategies

To identify an optimal synergy between Character LoRA and Style LoRA, AdaptiveFuse implements multiple Noise-based Fusion (Dynamic Adaptation) strategies.

1. **Noise Latents:** Compute a basic noise measure (e.g., sum of absolute values) for each LoRA' s predicted noise with various settings. Let $\epsilon_i(\mathbf{z}_t, t)$ denote the noise residual predicted by model $i \in \{q, r\}$ at timestep t . For example, define:

$$\text{noiseLatents}_q(t) = \sum |\epsilon_q(\mathbf{z}_t, t)|, \quad \text{noiseLatents}_r(t) = \sum |\epsilon_r(\mathbf{z}_t, t)|. \quad (3-2)$$

Then, a simple choice for the fusion weight α_t might be:

$$\alpha_t = \frac{\text{noiseLatents}_q(t)}{\text{noiseLatents}_q(t) + \text{noiseLatents}_r(t)}, \quad (3-3)$$

which proportionally favors the model with the larger noise magnitude.

2. **Noise Diff:** Assess noise changes over consecutive timesteps to ascertain which LoRA should dominate. Again, let $\epsilon_i(\mathbf{z}_t, t)$ be the noise predicted by model i . Define the “noise difference” measure:

$$\text{noiseDiff}_q(t) = \sum |\epsilon_q(\mathbf{z}_t, t) - \epsilon_q(\mathbf{z}_{t-1}, t-1)|, \quad (3-4)$$

$$\text{noiseDiff}_r(t) = \sum |\epsilon_r(\mathbf{z}_t, t) - \epsilon_r(\mathbf{z}_{t-1}, t-1)|. \quad (3-5)$$

A simple example of using this metric in the fusion weight could be:

$$\alpha_t = \frac{\text{noiseDiff}_q(t)}{\text{noiseDiff}_q(t) + \text{noiseDiff}_r(t)}, \quad (3-6)$$

biasing the fusion toward whichever model' s noise *changed* more from the previous step.

3. **Noise Ratio:** Track the ratio of noise change to guide dynamic updates of α_t . In this approach, we look at the relative change per timestep:

$$\text{noiseRatio}_q(t) = \frac{\sum |\epsilon_q(\mathbf{z}_t, t) - \epsilon_q(\mathbf{z}_{t-1}, t-1)|}{\sum |\epsilon_q(\mathbf{z}_{t-1}, t-1)|}, \quad (3-7)$$

$$\text{noiseRatio}_r(t) = \frac{\sum |\epsilon_r(\mathbf{z}_t, t) - \epsilon_r(\mathbf{z}_{t-1}, t-1)|}{\sum |\epsilon_r(\mathbf{z}_{t-1}, t-1)|}. \quad (3-8)$$

One could then construct α_t by comparing these two ratios, for instance:

$$\alpha_t = \frac{\text{noiseRatio}_q(t)}{\text{noiseRatio}_q(t) + \text{noiseRatio}_r(t)}. \quad (3-9)$$

This emphasizes the model whose noise residual exhibits the greater *relative* change over the last step.

Based on our ablation study in section 3.6.3, we found that the *Manual Tuning* approach consistently delivers a strong trade-off between preserving a subject’s identity and applying a desired artistic style. Consequently, the qualitative example (*Noise Fusion (Dynamic)*) in result section 3.6.1 illustrates results obtained with *Manual Tuning*, especially Manual (Style LoRA), alongside other comparison methods.

Algorithm 1 AdaptiveFuse Generation Process

Input: Pose P , text prompt T_p

Output: Generated output O

- 1: **Initialize:** Character LoRA-Net, Style LoRA-Net, and latent $\mathbf{z}_T$ from noise
 - 2: **for** $t = T$ **down to** 1 **do**
 - 3: $\epsilon_q(\mathbf{z}_t, t) \leftarrow \text{CharacterLoRA-Net}(\mathbf{z}_t, t, T_p, P)$
 - 4: $\epsilon_r(\mathbf{z}_t, t) \leftarrow \text{StyleLoRA-Net}(\mathbf{z}_t, t, T_p, P)$
 - 5: **Set** α_t (e.g., Noise Latents, Noise Ratio etc.)
 - 6: $\epsilon(\mathbf{z}_t, t) \leftarrow \alpha_t \times \epsilon_q(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_r(\mathbf{z}_t, t)$
 - 7: $\mathbf{z}_{t-1} \leftarrow \epsilon(\mathbf{z}_t, t)$
 - 8: **end for**
 - 9: **Decode** $\mathbf{z}_0$ to get output O
 - 10: **return** O
-

3.3 Theoretical Rationale

While the previous sections have introduced the *AdaptiveFuse* approach from an algorithmic standpoint, this section delves into the theoretical underpinnings of *noise-level fusion* and

explains why it can be more robust than parameter-space (weight) merging or simple module switching. In essence, noise residuals in diffusion models can be viewed as local gradient (score) estimates of the data distribution [2, 3], which makes their linear combination more meaningful than naively interpolating model parameters.

3.3.1 Noise Residuals as Gradient Estimates

In DDPMs and similar score-based frameworks, sampling proceeds by iteratively *removing* estimated noise from a noisy latent $\mathbf{x}_t$:

$$\mathbf{x}_{t-1} = \mathbf{x}_t - \epsilon_\theta(\mathbf{x}_t, t), \quad (3-10)$$

where $\epsilon_\theta(\mathbf{x}_t, t)$ is the predicted noise residual at step t . If we adopt a score-based perspective [3], then

$$\epsilon_\theta(\mathbf{x}_t, t) \approx -\nabla_{\mathbf{x}} \log p(\mathbf{x}_t),$$

meaning that the network's output can be seen as estimating the gradient of the log-likelihood. In this interpretation, subtracting the noise is akin to following a local gradient that pushes the sample toward higher-density regions. Consequently, *fusing* noise residuals from multiple models can be viewed as *combining* their individual gradient estimates.

3.3.2 Comparison with Parameter-Space Fusion

Challenges of Weight Merging. Parameter-space fusion, such as

$$\theta = \alpha \theta_A + (1 - \alpha) \theta_B, \quad (3-11)$$

often suffers from destructive interference [11, 12]. Modern diffusion models have millions (or billions) of parameters interconnected through U-Net layers, attention blocks, and residual connections. Linearly interpolating two distinct sets of weights θ_A and θ_B can place the fused model in a region of parameter space that is suboptimal especially when architectures or training distributions differ. Small conflicts in early layers can propagate into severe artifacts.

Why Noise-Level Fusion Helps. By contrast, *noise residuals* from different models usually have the same shape (e.g., the same (C, H, W) latent dimensions) and can be mixed *additively* without forcing any alignment in internal parameter structures. Concretely, at each timestep t :

$$\epsilon(\mathbf{x}_t, t) = \alpha_t \epsilon_q(\mathbf{x}_t, t) + (1 - \alpha_t) \epsilon_r(\mathbf{x}_t, t),$$

as given in Eq. (3-1) (Section 3.2.3). This operation simply combines gradient-like quantities (scores) rather than raw weights. If both models approximately learn the same underlying diffusion/likelihood framework, then their outputs lie in a *common noise space*, making linear blending far less prone to catastrophic interference.

3.3.3 A Simplified Multi-Objective View

Consider two models, A and B , each specialized for a different objective:

$$\epsilon_A(\mathbf{x}_t, t) \approx \nabla_{\mathbf{x}} J_A(\mathbf{x}_t), \quad \epsilon_B(\mathbf{x}_t, t) \approx \nabla_{\mathbf{x}} J_B(\mathbf{x}_t),$$

where J_A might represent an *identity-preservation* energy, while J_B enforces a particular *style* or other constraints. The fused noise

$$\epsilon(\mathbf{x}_t, t) = \nabla_{\mathbf{x}} \left[\alpha J_A(\mathbf{x}_t) + (1 - \alpha) J_B(\mathbf{x}_t) \right]$$

corresponds to the gradient of a *combined* energy function, $\alpha J_A + (1 - \alpha) J_B$. In principle, this allows us to trade off the two objectives in a well-defined way, without requiring that θ_A and θ_B be compatible at the parameter level. Hence, each model's specialized knowledge is preserved in the fused noise residual.

3.3.4 Architecture-Mismatch Robustness

A core advantage of fusing noise rather than weights is that each model's internal structure can differ. For instance, one LoRA might be trained on high-resolution face images with custom attention layers, while another focuses on a painterly style. As long as *both* final outputs (the predicted noise tensors) share a compatible shape, direct addition is feasible. Of course, if the models are trained on completely unrelated data distributions, or produce incompatible noise



scales, fusion might fail. Nevertheless, for closely related domains (e.g., human portraiture under stable diffusion), noise-level fusion is far more tolerant of architectural discrepancies than parameter merging [13, 14].

3.3.5 Noise Fusion as a Basis for Dynamic Scheduling

Because we combine noise residuals at each diffusion step, we can smoothly vary the blending ratio α_t over time. This property is unique to noise-level fusion, as parameter-space or “switch-based” approaches are typically forced to either (i) fix the interpolation or (ii) abruptly swap entire networks. In contrast, a schedule $\{\alpha_1, \alpha_2, \dots, \alpha_T\}$ allows us to place more emphasis on certain attributes (e.g., subject identity) early in the denoising process and shift toward others (e.g., stylization) later. Our ablation studies (Section 3.6.3) demonstrate how this dynamic control can achieve better trade-offs.

3.3.6 Summary of Theoretical Benefits

In short, noise-level fusion is theoretically appealing because:

- It treats each model’s output as a *score* or *gradient* in a *common noise space*, avoiding the destructive interference often seen in parameter-space interpolation.
- Different architectures are more easily accommodated, since only the output *tensor shape* must match.
- Multiple objectives (e.g., identity and style) can be combined in a manner analogous to adding their gradients, enabling multi-objective optimization without retraining.
- The method naturally supports *dynamic* scheduling of contributions per timestep, yielding finer-grained control than discrete module switching.

Thus, fusing at the noise level strikes a practical balance between flexibility and stability, particularly for tasks (like stylization + identity preservation) where direct weight merging easily leads to compromised performance. The following experimental results in Sections 3.5–3.6.1

substantiate these theoretical claims, demonstrating that noise-level fusion outperforms simpler baselines under multiple quantitative metrics.

3.4 Experimental Setup

3.4.1 Base Models

Our experiments utilize publicly available LoRA models in conjunction with stable diffusion backbones, as following LoRA Composite [12]. These LoRA models encompass high-resolution images annotated with character identities and accompanying artistic styles. Such diversity validates the breadth and robustness of the tested fusion strategies.

- **Character Identity:** We adopt LoRA models trained to reproduce the distinctive facial features of Scarlett Johansson [81], IU [82] and Dwayne ”The Rock” Johnson [83]. This serves as our Character LoRA for preserving identity.
- **Artistic Styles:** We incorporate style LoRAs for various painting or illustration genres, such as line art [84], oil painting [85], film color [86], brightness adjustment [87] and ink style [88]. Each style LoRA imposes unique texture, stroke patterns, or color schemes.

3.4.2 Baselines

To assess the efficacy of our proposed noise-fusion approach, we compare it against three alternative strategies that combine LoRA modules in more conventional ways:

- (1) **Weight Merging.** This method from ZipLoRA [11] merges parameter updates by averaging at the weight level.
- (2) **Switch-based Fusion.** This strategy from LoRA Composite [12] alternates between different LoRA modules at specific timesteps during the denoising process. For instance, early iterations might apply the Character LoRA to solidify identity, followed by later timesteps focused on the Style LoRA for stylistic rendering.

(3) **Noise Fusion (Half).** In this simpler setup, the noise predictions from Character LoRA and Style LoRA are combined with fixed, equal weights at each timestep:

$$\epsilon(\mathbf{z}_t, t) = 0.5 \times \epsilon_q(\mathbf{z}_t, t) + 0.5 \times \epsilon_r(\mathbf{z}_t, t). \quad (3-12)$$

Collectively, these baselines illustrate straightforward LoRA-combination techniques, each susceptible to issues like diluted identity, subdued stylistic cues, or abrupt transitions. In subsequent experiments, we show how dynamic noise-level fusion can overcome these pitfalls by adaptively blending LoRA outputs at each denoising step.

3.4.3 Implementation Details

- **Hardware:** All experiments run on a single NVIDIA 3090 GPU with 24 GB of memory.
- **Inference Steps:** We set the total number of denoising steps to 30 for each generated image, balancing generation quality and computational overhead.
- **Resolution:** We generate 768×1024 images (portrait orientation).
- **Pose Setting:** We employ Open Pose data from [10].
- **Pre-training:** We load publicly available stable diffusion models (e.g., Stable Diffusion 1.5) and then apply LoRA modules fine-tuned on specific identities or styles. No additional fine-tuning is performed beyond applying the LoRA.

3.5 Evaluation

We conduct both **qualitative** and **quantitative** evaluations, as following ZipLoRA [11]. Qualitatively, we rely on visual inspection to confirm that the fused outputs preserve the subject’s distinctive facial traits and integrate the target artistic style with minimal artifacts. To supplement these observations, we incorporate a quantitative evaluation that measures *Text CLIP*, *Image CLIP*, and *Face Distance* metrics for every generated image.

Quantitative Comparisons. Beyond visual inspection, we measure how well each fusion scheme matches textual prompts, reference style images, and a specific subject’ s face, using the script above (adapted to reflect our experiment details). In particular:

- **Text CLIP Score:** We use CLIP [46] to encode both the generated image and the text prompt, then compute the cosine similarity of the resulting embeddings. Higher similarity indicates better alignment with the textual description.
- **Image CLIP Score:** We compare the generated image against a style image, encoding both via CLIP and measuring their similarity. This metric helps assess how closely the final image adheres to the target style.
- **Face Similarity:** To verify identity preservation, we feed the face region from the generated image into face recognition [89] and compare the resulting embedding to that of the real subject’ s reference image. A lower face similarity indicates better faithfulness to the subject’ s identity, while a distance near 1.0 typically indicates poor identity retention.

3.6 Results

3.6.1 Qualitative Analysis

Figure 3.2 illustrates the fusion of three main inputs:

- **ControlNet pose** that dictates the subject’ s body arrangement,
- **Character LoRA** (Scarlett Johansson) tailored to preserve individual identity (facial structure and other personal traits),
- **Style LoRA** responsible for rendering a distinct artistic look (e.g., ink drawing or oil painting).



Figure 3.2 Qualitative demonstration of fusing Character LoRA and Style LoRA with ControlNet. From left to right: (1) Merge, (2) Switch, (3) Noise Fusion (Half), and (4) Noise Fusion (Dynamic). Unlike previous approaches, our noise-based method successfully retains identity and integrates style.

We compare four distinct methods: *Merge*, *Switch*, *Noise Fusion (Half)*, and *Noise Fusion (Dynamic)*. The first three highlight common limitations of weight-centric or fixed-fraction fusion, such as partly losing the subject’s identity or muting certain style aspects. In contrast, the final column (*Noise Fusion (Dynamic)*) demonstrates how our noise-based integration strategy adaptively balances the two LoRAs’ contributions: identity details (like the character’s facial shape) remain sharply defined, while the stylistic traits (brushstrokes, color schemes) appear more vivid. As a result, the Noise Fusion (Dynamic) method avoids abrupt inconsistencies in hair color and facial structure and yields more coherent and visually appealing images.

Key Observations:

- *Identity Preservation:* Using Character LoRA ensures the subject’s distinctive features (facial structure, hair color) remain consistent.
- *Style Integration:* The Style LoRA imposes unique brushstrokes and color palettes, which

remain intact due to noise-based fusion rather than destructive weight merges.

- *Overall Quality*: Compared to conventional fusion strategies, AdaptiveFuse yields sharper, more detailed results with fewer artifacts.

3.6.2 Quantitative Comparisons

To complement our qualitative observations, we evaluate three numerical metrics for each of the four fusion approaches (*Merge*, *Switch*, *Noise Fusion (Half)*, *Noise Fusion (Dynamic)*):

Table 3.1 summarizes our results. Note that whereas *Text CLIP* and *Image CLIP* should be as *large* as possible, *Face Similarity* values should be as *low* as possible to signify minimal deviation from the subject’s true facial features.

Table 3.1 Comparison of LoRA fusion strategies under three metrics. *Text CLIP* and *Image CLIP* scores are higher-is-better; *Face Similarity* (distance) is lower-is-better. Bold indicates the best outcome per metric.

Method	Text CLIP (↑)	Image CLIP (↑)	Face Similarity (↓)
Merge	0.2580	0.6255	0.5661
Switch	0.2659	0.5949	0.4831
Noise Fusion (Half)	0.2682	0.5941	0.4989
Noise Fusion (Dynamic)	0.2722	0.5613	0.4781

Analysis.

- **Text CLIP (Higher is Better)**. The *Noise Fusion (Dynamic)* strategy surpasses all others (0.2722), confirming that fine-grained control over fusion weights can yield stronger alignment with the text prompt.
- **Image CLIP (Higher is Better)**. *Merge* excels at style adherence (0.6255), suggesting that blending LoRA parameters without restraint can amplify stylistic details. However, this often comes at the expense of identity fidelity.

- **Face Similarity (Lower is Better).** The *Noise Fusion (Dynamic)* method achieves the smallest face similarity (0.4781), indicating better retention of the subject’s facial features compared to *Merge* (0.5661) or *Switch* (0.4831). By carefully balancing LoRA contributions, *Noise Fusion (Dynamic)* mitigates identity distortion.
- **Trade-off Insights.** If style strength is paramount (e.g., vivid painterly effects), *Merge* provides the highest Image CLIP but suffers in facial resemblance. Conversely, *Noise Fusion (Dynamic)* offers robust identity retention yet yields a milder style effect. *Switch* and *Noise Fusion (Half)* occupy a middle ground. Ultimately, noise-level fusion is versatile, allowing users to prioritize different objectives by tuning α_t or employing specialized schedules.

3.6.3 Ablation Study

In addition to the previously discussed fusion approaches, and various *Noise-based* schedules), we also compare the cases of using *only the Character LoRA* or *only the Style LoRA* (i.e., a single LoRA-driven diffusion U-Net without fusion). Furthermore, we perform an ablation study using two different *manual* schedules to control α_t across denoising timesteps. Specifically:

- **Manual (Style LoRA):** We gradually increase α_t from 0 to 0.6 over the denoising steps, giving more weight to the *style* noise residual:

$$\alpha_t \in \{0, 0, \dots, 0.6, 0.6\}, \quad (3-13)$$

$$\epsilon(\mathbf{z}_t, t) = \alpha_t \times \epsilon_r(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_q(\mathbf{z}_t, t). \quad (3-14)$$

- **Manual (Character LoRA):** Here, α_t also increases from 0 to 0.6, but it now emphasizes the *character* noise residual:

$$\alpha_t \in \{0, 0, \dots, 0.6, 0.6\}, \quad (3-15)$$

$$\epsilon(\mathbf{z}_t, t) = \alpha_t \times \epsilon_q(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_r(\mathbf{z}_t, t). \quad (3-16)$$

Figure 3.3 from our analyses shows how the scheduling of fusion weights evolves over timesteps, showcasing AdaptiveFuse’ s capacity to allocate greater or lesser influence to each LoRA as required. In addition to methods which we show at 3.6.1, we include methods such as Noise Latents, Noise Diff, Noise Ratio, and Manual Tuning.

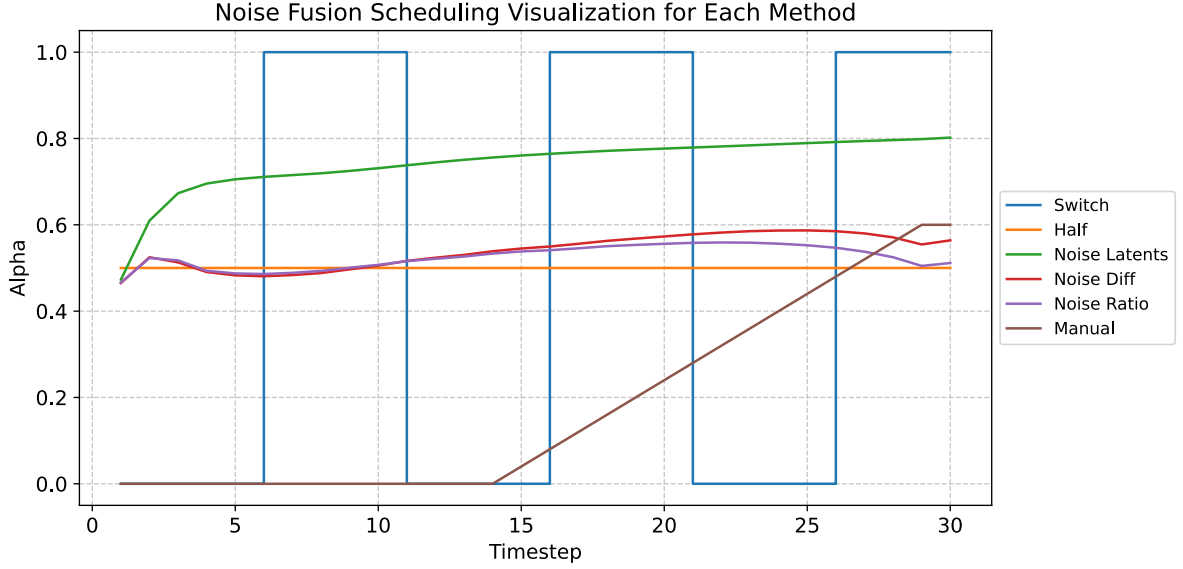


Figure 3.3 Visualization of fusion weights across denoising steps for each scheduling method. Different schedules allow for dynamic emphasis of character identity or stylistic cues.

Table 3.2 reports the quantitative performance for each of these strategies (including the single-LoRA baselines). We evaluate three metrics: *Text CLIP* ($\uparrow$), *Face Similarity* ($\downarrow$), and *Image CLIP* ($\uparrow$).

Observations.

- *Single-LoRA Baselines*: Only Character LoRA yields moderate textual alignment (Text CLIP = 0.2713), moderate style similarity (Image CLIP = 0.5605), and the best identity retention (Face Similarity = 0.4562). Only Style LoRA achieves lower text alignment (0.2495) but stronger style similarity (0.6355), at the cost of significantly worse identity retention (0.5930). These baselines demonstrate that relying solely on one LoRA compromises the other objective, underscoring the need to combine both.
- *Text CLIP* ($\uparrow$): The *Noise Ratio* and *Manual (Style LoRA)* methods achieve higher textual

Table 3.2 Ablation study results comparing nine fusion strategies (transposed). The top two rows show single-LoRA baselines, while the subsequent rows present fusion methods.

Method	Text CLIP ($\uparrow$)	Image CLIP ($\uparrow$)	Face Similarity ($\downarrow$)
Only Character LoRA	0.2713	0.5605	0.4562
Only Style LoRA	0.2495	0.6355	0.5930
Noise Latents	0.2697	0.5821	0.4827
Noise Diff	0.2704	0.5830	0.4761
Noise Ratio	0.2720	0.5798	0.4818
Manual (Character LoRA)	0.2549	0.6287	0.5459
Manual (Style LoRA)	0.2722	0.5613	0.4781

alignment (0.2720 and 0.2722, respectively). This indicates that carefully modulating α_t can enhance the semantic fidelity of generated images.

- *Image CLIP ($\uparrow$): Manual (Character LoRA)* attains the highest style similarity (0.6287), suggesting that front-loading the style influence can effectively preserve stylistic traits. However, this emphasis on style contributes to a relatively high Face Similarity (0.549).
- *Face Similarity ($\downarrow$): Noise Diff* attains the lowest face distance (0.4761), implying strong identity retention. *Noise Latents* is a close contender at 0.4827, and *Manual (Style LoRA)* also maintains a competitive ID distance (0.4781).

The two manual schedules (*Style LoRA* vs. *Character LoRA*) highlight how the rate of shifting weights strongly affects identity and style adherence. Meanwhile, *Noise-based* strategies often strike a moderate balance, automatically adapting α_t based on residual magnitudes or changes. This automatic adaptation can reduce the need for meticulous hand-tuning in practice.

Figure Figure 3.4. compares how different LoRA-based strategies (including two single-LoRA baselines and several fusion methods) affect the final generated images under two stylistic scenarios: an oil-painting look (top row) and a more photorealistic, brighter look (bottom row).



Figure 3.4 Qualitative results of the ablation study comparing single-LoRA baselines (*Only Character LoRA*, *Only Style LoRA*) and various fusion strategies (*Noise Latents*, *Noise Diff*, *Noise Ratio*, *Manual (Character LoRA)*, and *Manual (Style LoRA)*). The top row illustrates an oil-painting style scenario, while the bottom row shows a more photorealistic setting which makes a photo brighter.

Single-LoRA Baselines (*Only Character LoRA*, *Only Style LoRA*): Each baseline shows what happens when the diffusion model applies *only* a LoRA specialized for identity (*Character LoRA*) or *only* a LoRA specialized for style (*Style LoRA*). In the oil-painting row, *Only Style LoRA* yields vibrant painterly strokes yet loses some facial identity; conversely, *Only Character LoRA* retains the subject’s features but lacks an oil-painting aesthetic.

A similar trade-off appears in the photorealistic row: *Only Style LoRA* captures bright lighting but dilutes personal features, while *Only Character LoRA* keeps the subject’s face intact at the cost of a plainer style.

Fusion Strategies (*Noise Latents*, *Noise Diff*, *Noise Ratio*, *Manual (Character LoRA)*, and *Manual (Style LoRA)*): These columns illustrate how combining the two LoRAs (identity and style) can preserve essential facial details while still integrating the chosen art or photographic style.

Noise Latents, Noise Diff, Noise Ratio: These noise-based methods adaptively weigh each LoRA’s noise predictions at different denoising timesteps. They often yield a smoother combination of style and identity; notice, for instance, that the subject’s face remains consistent while brushstrokes or bright-lighting effects are applied more evenly.

Manual (Character LoRA), Manual (Style LoRA): In these approaches, a human custom-



tunes how the blend parameter α (i.e., the weight on Character vs. Style LoRA) shifts over time. Manually prioritizing the *Style LoRA* at later timesteps often boosts artistic traits, while prioritizing the *Character LoRA* helps lock in identity details before bringing in style. The Manual approaches enable the diverse output which aligns style and identity.

Overall, these results reinforce that noise-level fusion methods, especially properly tuned manual schedules, can yield more favorable trade-offs between identity preservation and style fidelity compared to single-LoRA baselines or simpler merging strategies (*Switch*, *Half*).

3.7 Discussion

3.7.1 Parameter Tuning

From the ablation results, the Manual schedules can outperform other methods *on certain metrics*, but do not consistently beat every other strategy across *all* metrics. For instance, Manual (Style LoRA) achieves a high Text CLIP score and respectable identity preservation, yet it yields a somewhat lower Image CLIP score. On the other hand, Manual (Character LoRA) excels at Image CLIP score but partially sacrifices identity fidelity. Meanwhile, some other noise-based approaches (e.g., *Noise Latents*, *Noise Diff*) are more automated, can strike a balanced trade-off, and in some cases achieve lower identity distance. So whether a manual schedule is better really depends on which metric; style fidelity, identity preservation, or textual alignment, is most important for your specific use case.

In essence, the "optimal" choice is contingent upon the user's priorities. If we want strong stylistic emphasis at the risk of losing some face details, we might manually weight the Style LoRA more heavily (*Manual (Style LoRA)*). If identity fidelity is paramount, we might prefer a noise-based method that adaptively balances both LoRAs; potentially without requiring constant human intervention. Hence, we *cannot* categorically say the manual setting is "much better" in all scenarios. Rather, it is one flexible option, especially if you are willing to tune α_t step by step and know precisely when (or how much) to emphasize either the character or the style.



3.7.2 Global Space Applicability

AdaptiveFuse’ s strong performance stems from isolating and fusing each LoRA’ s *noise residuals* rather than their raw parameters. This design drastically reduces destructive interference between style-specific and identity-specific updates. Our method is adaptable to more complex scenes that include additional objects, backgrounds, or specialized LoRAs (e.g., LoRA modules that emphasize lighting or composition). Manual or heuristic-based scheduling of α_t further allows fine-grained control to ensure no single LoRA dominates prematurely.

3.7.3 Theoretical Underpinnings of AdaptiveFuse

The intuition under the hood is different personalization controls (by lora) have different strengths for different timesteps and different layers. It is hard to control with model parameter fusion but easy in the noise domain. The empirical comparison to model fusion demonstrates noise fusion is more scalable and flexible. The proposed method itself could serve as an explanation tool for this effect. As shown in our ablations, changing the noise scales for different models at different stages affects the layout/appearances accordingly.

3.7.4 Limitations

While our experiments demonstrate that AdaptiveFuse can seamlessly combine LoRAs for style and identity *within the Stable Diffusion ecosystem* (including ControlNet and LoRA modules), the method’ s generality for other architectures or drastically different scheduling schemes is less explored. Further tests would be needed to confirm broad compatibility—an area we partially address in Chapter 4 by fusing an image-centric identity model with a motion-centric video diffusion model.

3.8 Summary

We introduced **AdaptiveFuse**, a training-free, noise-based fusion framework that merges a *Character LoRA* and a *Style LoRA* within a single generative pipeline. By manipulating the noise residuals at inference time, our approach not only achieves better style/identity synergy



than conventional weight-based merges, but also provides flexible scheduling to accommodate varying user or domain preferences. Quantitative evaluations confirm that *Noise Latents* excels at preserving the subject's facial identity while accurately delivering the intended artistic style, yielding the highest Image CLIP and Face Similarity. These promising results set the stage for Chapter 4, where we extend the AdaptiveFuse concept to tackle the more challenging domain of dancing video generation, emphasizing temporal coherence alongside identity fidelity.



Chapter 4. Fusion for Dancing Video Generation

Building on the success of Character LoRA and Style LoRA Fusion, this chapter presents a novel application of the AdaptiveFuse framework to the challenging task of dancing video generation. By seamlessly integrating models specialized in global motion dynamics with those excelling in preserving fine-grained facial details, AdaptiveFuse addresses long-standing issues in consistency and realism. The below sections detail the motivation, method, and evaluate the performance of our approach. Through adaptive noise-based fusion, we demonstrate significant improvements in both temporal coherence and identity preservation, pushing the boundaries of high-quality, personalized dancing video synthesis.

4.1 Motivation

Generating high-quality dancing videos that preserve fine-grained facial details and expressions is a multifaceted challenge. This task necessitates the integration of multiple specialized capabilities to ensure both temporal dynamics and identity consistency. Existing diffusion-based animation methods, while proficient in capturing overall motion and appearance, often falter in maintaining consistent and detailed facial features. Notably, methods like *DisCo* [24] and *MagicAnimate* [23] exhibit inconsistencies and artifacts in facial regions, undermining the realism and identity preservation of the generated videos.

In order to work on the limitations, it is imperative to implement an approach that integrates specialized models with distinct focuses on various aspects of video generation. Specifically, combining a model adept at capturing temporal motion with another specialized in refining facial details can enhance the quality and realism of dancing videos. This fusion aims to take advantage of the strengths inherent in each model, thereby ensuring both dynamic motion and high-fidelity facial features in the final output.

4.2 Methodology

This section introduces the AdaptiveFuse framework, designed to enhance dancing video generation by seamlessly integrating multiple pre-trained diffusion models. The framework leverages specialized models to capture both overall motion dynamics and fine-grained facial details, ensuring high-quality and identity-consistent video outputs.

4.2.1 AdaptiveFuse Framework for Dancing Video Generation

Building upon the success of AdaptiveFuse in image generation, this work extends the framework to the domain of dancing video generation. The approach involves fusing two key components:

- **AnimateAnyone-Net:** A cutting-edge dancing video generation framework that excels in capturing overall movement and appearance based on reference images and pose sequences [10].
- **Face-Net:** A specialized module designed to refine facial features and expressions, ensuring high-fidelity and identity-consistent face generation. Our Face-Net module is built by combining IP-Adapter FaceID [21] and ControlNet [20] to refine the facial region.

In order to more clearly discern the complementary strengths of each component, we also visualize the video generation results obtained using only Face-Net and only AnimateAnyone-Net in Figure 4.1. Although Face-Net encounters challenges in generating clothing and backgrounds, it specializes in generating high-quality facial details, capturing the likeness and expressions of the reference face. In contrast, AnimateAnyone-Net focuses on the overall motion and appearance of the character but may struggle with fine-grained facial features. The complementary strengths of these two networks highlight the importance of employing an adaptive fusion strategy to effectively combine their outputs.

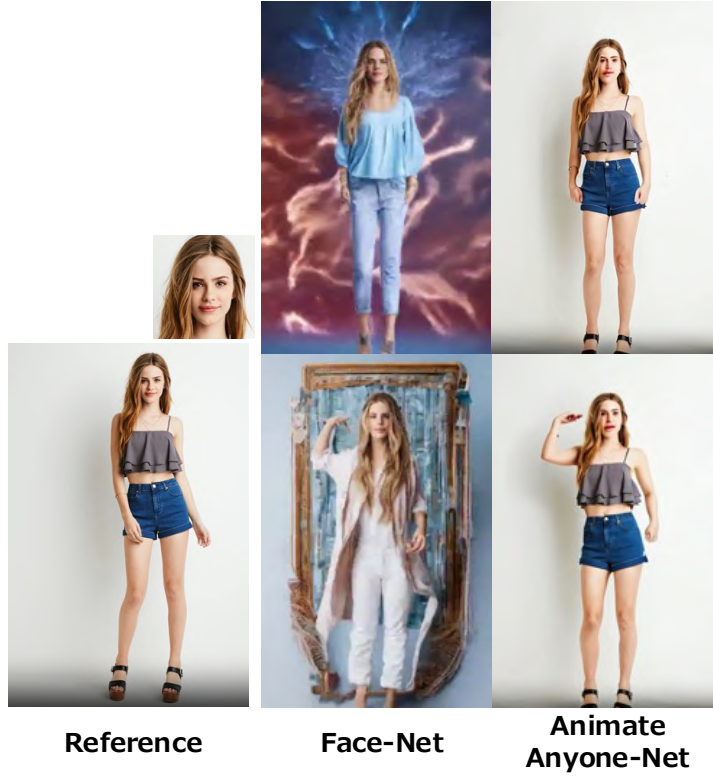


Figure 4.1 Visualization of the video generation results obtained using only Face-Net and AnimateAnyone-Net. Face-Net specializes in generating high-quality facial details, while AnimateAnyone-Net focuses on the overall motion and appearance of the character. The complementary strengths of these networks underscore the importance of adaptive fusion for optimal results.

4.2.2 Fusion Module

The Fusion Module is the core component that integrates the outputs of AnimateAnyone-Net and Face-Net. It employs a mask-based approach to combine the refined facial features from Face-Net with the body and background generated by AnimateAnyone-Net. The fusion process is governed by the noise residuals computed at each denoising step, ensuring a smooth and coherent integration.

Mathematically, the fusion process is described as:

$$\epsilon(\mathbf{z}_t, t) = M \times [\alpha_t \times \epsilon_q(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_r(\mathbf{z}_t, t)] + (1 - M) \times \epsilon_q(\mathbf{z}_t, t) \quad (4-1)$$

where:

- $\epsilon(\mathbf{z}_t, t)$ is the fused noise residual at timestep t .
- $\epsilon_q(\mathbf{z}_t, t)$ is the noise residual output from AnimateAnyone-Net.

- $\epsilon_r(\mathbf{z}_t, t)$ is the noise residual output from Face-Net.
- M is the face mask identifying the face area.
- α_t is the fusion parameter controlling the influence of each network at timestep t .

4.2.3 Face-Net Integration

Face-Net integrates *IP-Adapter FaceID* [21] and *ControlNet* [20] to enhance facial feature generation. *IP-Adapter FaceID* extracts identity-specific information from the reference image. This ensures that the output faces preserve the unique traits of the subject. *ControlNet* conditions the face generation on desired poses using face masks, enabling dynamic expressiveness and consistency with the intended motion.

This dual approach ensures that the generated faces are both identity-preserving and dynamically expressive, addressing the limitations observed in existing methods where facial details are often inconsistent or distorted.

4.2.4 Noise-based Fusion Schedulers

To address discrepancies in noise schedules between AnimateAnyone-Net and Face-Net, AdaptiveFuse employs various fusion schedulers that adapt the fusion parameters based on noise levels, as described in Section 3.2.3 Fusion Strategies.

4.2.5 Algorithmic Overview

Figure 4.2 introduces the AdaptiveFuse framework, illustrating the interaction between AnimateAnyone-Net, Face-Net, and the Fusion Module. The algorithmic process is summarized in Algorithm 2.

**Algorithm 2** AdaptiveFuse Dancing Video Generation**Input:** Reference image I , pose sequence $\{P_0, \dots, P_{n-1}\}$, face mask M , text prompt T_p **Output:** Generated video V

- 1: **Initialize:** AnimateAnyone-Net, Face-Net, and latent $\mathbf{z}_T$ from noise
- 2: **for** $t = T$ **down to** 1 **do**
- 3: $\epsilon_q(\mathbf{z}_t, t) \leftarrow \text{AnimateAnyone-Net}(\mathbf{z}_t, t, I, P_t, T_p)$
- 4: $\epsilon_r(\mathbf{z}_t, t) \leftarrow \text{Face-Net}(\mathbf{z}_t, t, I, P_t, T_p)$
- 5: **Compute** α_t (via scheduler or manual)
- 6: **Fuse noise residual:**

$$\epsilon(\mathbf{z}_t, t) = M \times [\alpha_t \times \epsilon_q(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_r(\mathbf{z}_t, t)] + (1 - M) \times \epsilon_q(\mathbf{z}_t, t)$$

- 7: $\mathbf{z}_{t-1} \leftarrow \epsilon(\mathbf{z}_t, t)$
- 8: **end for**
- 9: $V \leftarrow \text{Decode}(\mathbf{z}_0)$
- 10: **return** V

4.3 Experimental Setup

4.3.1 Datasets

In order to assess the robustness, as following previous methods [10, 24, 25, 39], we employ two benchmark datasets:

- **TikTok Dataset** [90]: Comprises approximately 350 single-person dance videos, each spanning 10-15 seconds, offering diverse subjects and motions.
- **UBC Dataset** [91]: Contains a total of 600 videos, depicting various actions such as standing, moving sideways, and turning around, providing a broad range of motion dynamics.

For each dataset, we employ test datasets, and the initial frame of each video serves as

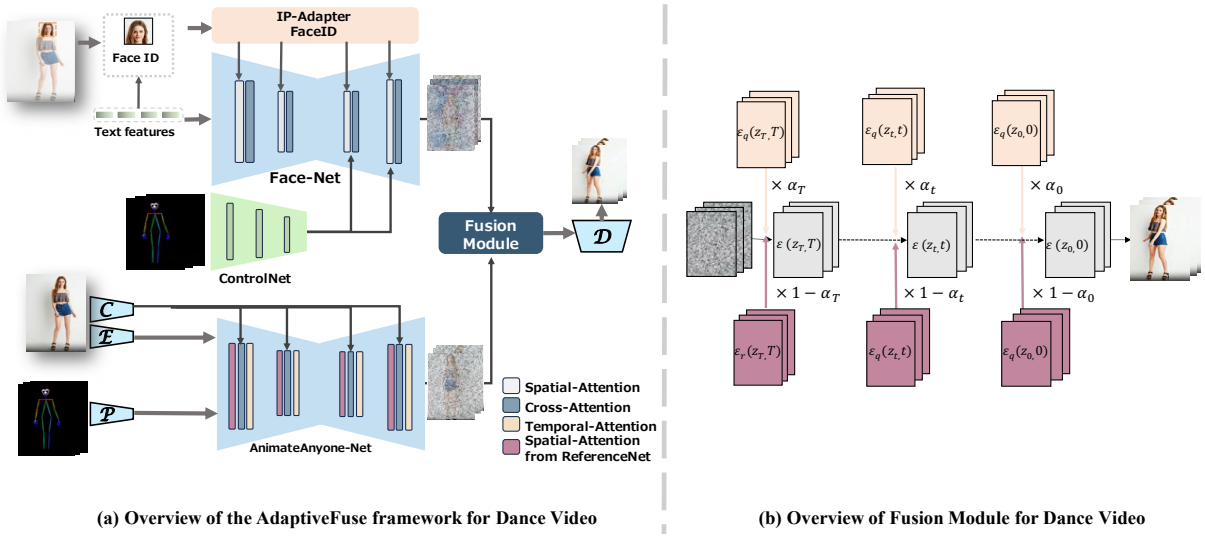


Figure 4.2 Overview of the AdaptiveFuse framework for Dance Video Generation. (a) **Architecture of AdaptiveFuse framework for Dance Video Generation.** For Face-Net, the face image is extracted using InsightFace, and *IP-Adapter FaceID* extracts face features. *ControlNet OpenPose* encodes the pose sequence. For AnimateAnyone-Net, the reference image is encoded by the VAE Encoder and CLIP to extract appearance features. The Pose Guider encodes the pose sequence. The encodings are respectively input to AnimateAnyone-Net and Face-Net, which contain U-Nets that iteratively compute noise residuals for the video frames. The Fusion Module uses the noise residuals and fuses them by applying weights with a face mask. The denoised frames are finally decoded by the VAE Decoder into the video clip. (b) **Overview of Fusion Module for Dance Video Generation.** The Fusion Module dynamically combines the outputs of AnimateAnyone-Net and Face-Net at the noise level. The mask-based fusion mechanism ensures that the facial region is predominantly refined by Face-Net, while the body and background details are derived from AnimateAnyone-Net. The fusion process follows Equation 4-1, where α controls the contribution of each model at timestep t .

the reference image, while the subsequent frames are extracted to form a pose sequence representing the dynamic movement. Pose estimation is performed using DWPose [92], and facial identity information is extracted using InsightFace [93]. The facer tool [94] is utilized for mask extraction to accurately identify facial regions.

4.3.2 Implementation Details

The AdaptiveFuse framework is deployed with PyTorch and executed on a single NVIDIA 3090 GPU with 24GB of memory. Generated videos have a resolution of 512×512 pixels and consist of 24 frames, processed over 30 denoising steps to balance quality and computational efficiency. Pre-trained models, including AnimateAnyone-Net and Face-Net, are utilized without further fine-tuning, emphasizing the training-free nature of the fusion approach.



4.3.3 Evaluation Metrics

We evaluate the proposed method by employing a combination of quantitative and qualitative metrics:

- **Quantitative Metrics:**

- **L1 Error:** Evaluates pixel-wise deviations between generated frames and ground-truth frames.
- **PSNR (Peak Signal-to-Noise Ratio):** Evaluates image quality based on pixel intensity differences [95].
- **SSIM (Structural Similarity Index):** Gauges how closely the generated and reference frames align in terms of structural content [96].
- **LPIPS (Learned Perceptual Image Patch Similarity):** Quantifies perceptual resemblance by comparing deep feature representations [97].
- **FID (Fréchet Inception Distance):** Measures the alignment between the distributions of generated and real images [98].
- **FVD (Fréchet Video Distance):** Assesses the temporal consistency and realism of generated videos [99].
- **Adaface Score:** Computes cosine similarity between features of generated and reference faces to evaluate identity preservation [100].
- **CLIP Score (Image CLIP Score):** Measures alignment between generated videos and corresponding image input [46].
- **Consistency Score:** Quantifies temporal stability and smoothness of motion across frames [101].

- **Qualitative Evaluation:**

- Evaluates the realism and coherence of generated videos, focusing on facial detail preservation and motion fidelity.

4.3.4 Baselines

The proposed method is compared against various baselines:

1. **DisCo** [24]: Employs a diffusion model supplemented by disentangled conditioning modules for pose, human appearance, and background, targeting human image animation.
2. **MagicAnimate** [23]: Leverages a video diffusion framework and an appearance encoder to capture temporal aspects and maintain identity, using DensePose motion cues and reference images.
3. **MagicPose** [25]: Composes a multi-source attention module providing detailed appearance guidance for human-centric video generation.
4. **AnimateAnyone** [10]: A pre-trained human image animation framework that generates consistent dancing videos from reference images and pose sequences but may lack fine-grained facial detail preservation.

4.4 Results

4.4.1 Quantitative Results

Table 4.1 displays a metric-based evaluation between our approach and the cutting-edge baselines across various image quality, video quality, and consistency metrics. Our approach excels in preserving facial identity and generating realistic faces, as evidenced by the highest Adaface score of 0.390, highlighting the effectiveness of our Face-Net module in refining facial features and maintaining consistency with the reference image.

Regarding image quality, our method obtains the highest SSIM of 0.805 and the lowest L1 error of $4.94e-05$, indicating excellent structural similarity and pixel-level accuracy. We also achieve a PSNR of 19.4, matching the top-performing AnimateAnyone. The LPIPS score of 0.153 is comparable to AnimateAnyone’s 0.150, suggesting similar perceptual quality.

For video quality, our method exhibits the lowest FVD of 13.1, slightly surpassing AnimateAnyone’s 13.2, indicating stronger temporal continuity and visual fidelity. The FID score

of 24.7 is the second lowest, closely following AnimateAnyone’s 23.3, indicating the realistic nature of the synthesized video frames.

Regarding consistency and faithfulness to the input, our method achieves the highest CLIP score of 0.900, showcasing a strong consistency among the output videos and their corresponding text descriptions. Additionally, our consistency score of 0.952 is on par with AnimateAnyone’s 0.953, reflecting the temporal stability of the generated videos.

Notably, our approach significantly outperforms other baselines, including DisCo, MagicAnimate, and MagicPose, in terms of facial identity preservation and overall face quality. DisCo and MagicAnimate exhibit lower Adaface scores of 0.115 and 0.274, respectively, indicating their limitations in generating consistent and realistic faces. MagicPose achieves a higher Adaface score of 0.375 but still falls short of our method’s performance.

The superior Adaface score achieved by our method underscores the efficacy of our Face-Net module and the adaptive fusion technique. By dedicating a specialized network to refine facial features and seamlessly integrating its output with the content generated by AnimateAnyone, our approach ensures high-quality and identity-consistent face generation in the resulting videos.

Overall, the quantitative findings establish the leading performance of our approach in generating realistic and consistent human dance videos, with a particular emphasis on enhancing facial quality and preserving identity. The combination of the Face-Net module and the adaptive fusion strategy enables our approach to outperform existing methods in the critical aspect of face generation while maintaining competitive results in other evaluation metrics.

Table 4.1 Quantitative comparisons of our method with recent SOTA methods. $\uparrow$ means higher is better, and $\downarrow$ means lower is better. Bold numbers indicate the best result in each metric.

Metric	DisCo	MagicAnimate	MagicPose	AnimateAnyone	Our Method
PSNR $\uparrow$	16.6	7.86	13.7	19.4	19.4
SSIM $\uparrow$	0.657	0.524	0.689	0.797	0.805
L1 $\downarrow$	7.75e-05	33.1e-04	16.2e-04	5.06e-05	4.94e-05
LPIPS $\downarrow$	0.235	0.457	0.323	0.150	0.153
FID $\downarrow$	57.4	62.2	37.1	23.3	24.7
FVD $\downarrow$	53.4	83.6	71.5	13.2	13.1
Consistency $\uparrow$	0.912	0.943	0.924	0.953	0.952
CLIP $\uparrow$	0.831	0.829	0.854	0.899	0.900
Adaface $\uparrow$	0.115	0.274	0.375	0.354	0.390

4.4.2 Qualitative Analysis

Figure 4.3 contrasts frames from dancing videos created by our approach (*far-right column*) with outputs from four baselines (DisCo, MagicAnimate, MagicPose, and AnimateAnyone). Each row corresponds to a different *reference image* (leftmost photos) and *pose* (OpenPose keypoints shown in the second column from the left). The reference image determines the subject’s identity and appearance, while the reference pose constrains the target motion to be replicated.

Columns under each baseline illustrate how well the generated frames preserve both global motion (matching the reference pose) and local features (notably the face region). Specifically, we highlight two primary aspects:

Facial Preservation. In Figure 4.3, we compare frames from the same dance sequence generated by each method:

- **DisCo, MagicAnimate:** Frequent face distortions, especially under large pose changes.
- **MagicPose:** Better alignment than DisCo, but subtle identity drift over time.
- **AnimateAnyone** (alone): Strong motion but occasional facial blurriness or mismatch in eyes/mouth.
- **Ours:** Retains sharper facial contours (thanks to Face-Net) while preserving motion continuity.

Temporal Coherence. Our approach yields fewer flickers in face shape or expression across adjacent frames. AnimateAnyone alone is also consistent in global motion but can lose local facial detail. Noise-level fusion addresses this shortcoming, bridging large movements with consistent facial identity.

Overall, the proposed method excels at preserving the subject’s identity in the face region while accurately transferring the pose from the reference skeleton, leading to smoother and more realistic dancing videos.



Figure 4.3 Comparative snapshots from videos generated using our method and other baseline methods, with a focus on the face region, demonstrating superior facial detail and temporal coherence in our approach.

Additional Qualitative Comparisons (UBC and TikTok).

Figures 4.4 and 4.5 showcase side-by-side video generation temporal snapshots for reference images from the UBC and TikTok datasets, respectively. Each figure is arranged in rows for different methods (DisCo, MagicAnimate, MagicPose, AnimateAnyone, and ours), highlighting the reference subject, the OpenPose-based *pose* sequence, and the resulting synthesized frames.

- **Reference Image and Pose:** The top rows display the subject’s identity (face, outfit) and the target body movements extracted via pose estimation.
- **Baselines:** DisCo, MagicAnimate, MagicPose, and AnimateAnyone each generate dancing sequences aligned with the reference pose but often introduce blurred facial features or frame-to-frame flickers.
- **Ours:** Our method (bottom rows) preserves sharper facial details and achieves smoother transitions, effectively retaining both identity and motion consistency. Notably, facial expressions remain more stable over time, greatly enhancing realism for dancing videos.

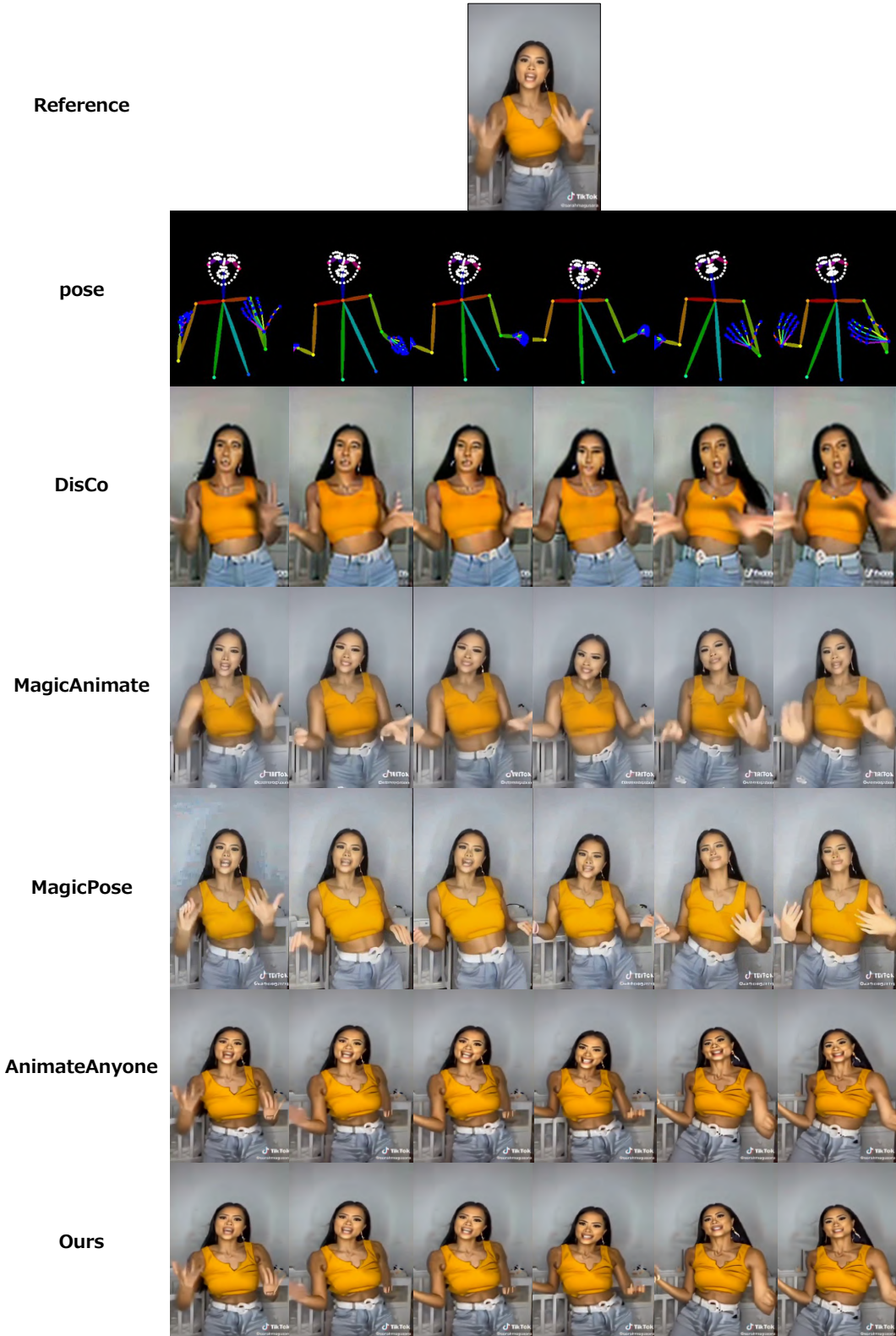


Figure 4.4 Qualitative comparison of video generation results on a reference image from the UBC dataset. We present snapshots from videos generated by our method and baselines. Our approach demonstrates superior facial detail preservation and temporal coherence, particularly evident in the consistent facial expressions and smooth transitions between frames.

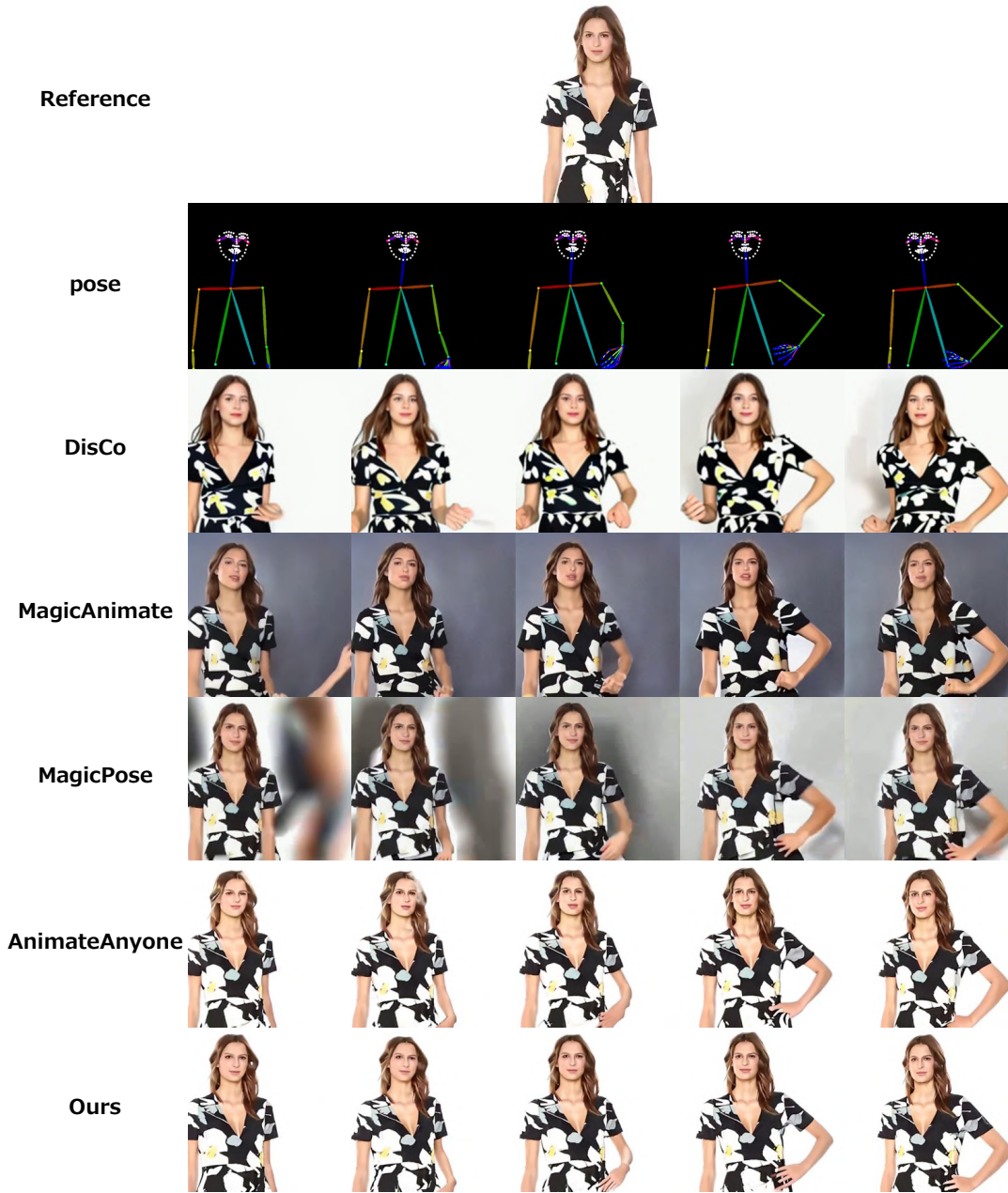


Figure 4.5 Qualitative comparison of video generation results on a reference image from the TikTok dataset. The figure illustrates frame sequences from videos generated by our method and competing approaches. Our method exhibits enhanced facial detail retention and improved temporal stability, evidenced by the consistent facial features and fluid motion across consecutive frames.

Overall, the comparisons confirm that our approach outperforms baselines on both datasets, delivering superior facial fidelity and temporal coherence in dynamic motion settings.

4.4.3 Ablation Study

In order to explore how different fusion parameters affect outcomes, we compare various fusion schemes, such as switch-based, half-weight, noise-based, and manual tuning strategies against a baseline without fusion. We focus on how well each strategy preserves facial details and overall identity, especially under challenging motion or pose transformations. Table 4.2 and Figures 4.6 and 4.7 showcase both the quantitative outcomes (e.g., PSNR, SSIM, Adaface) and qualitative snapshots.

Table 4.2 Quantitative ablation results. $\uparrow$ means higher is better, while $\downarrow$ means lower is better. **Bold highlights the best in each category.**

Metric	Without Fusing	Switch	Half	Noise Latents	Noise Diff	Noise Ratio	Manual
PSNR $\uparrow$	19.4	19.5	19.5	19.4	19.5	19.5	19.4
SSIM $\uparrow$	0.797	0.805	0.805	0.801	0.805	0.804	0.805
L1 $\downarrow$	5.06e-05	4.94e-05	4.93e-05	5.00e-05	4.94e-05	4.94e-05	4.94e-05
LPIPS $\downarrow$	0.150	0.152	0.151	0.153	0.152	0.152	0.153
FID $\downarrow$	23.3	24.5	24.5	24.8	24.6	24.6	24.7
FVD $\downarrow$	13.2	13.2	13.0	13.1	13.0	13.1	13.1
Consistency $\uparrow$	0.953	0.951	0.952	0.952	0.952	0.952	0.952
CLIP $\uparrow$	0.899	0.898	0.897	0.898	0.898	0.898	0.900
Adaface $\uparrow$	0.355	0.355	0.378	0.371	0.380	0.387	0.390

4.4.3.1 Quantitative Analysis

Table 4.2 presents the metrics for each strategy. Removing the fusion module (*Without Fusing*) decreases the Adaface score from 0.390 to 0.355, confirming the importance of fusion for high-fidelity face generation. **Manual Tuning** attains the highest Adaface score (0.390), underscoring the effectiveness of human guidance. Noise-based approaches (e.g., **Noise Diff**, **Noise Ratio**) also exceed the baseline but remain slightly below **Manual** and **Noise Latents**.

Overall, these ablation results demonstrate that noise-level integration significantly enhances identity consistency and visual quality, with both user-controlled (*Manual*) and automated (*Noise Latents*) schedules excelling in stabilizing facial features across diverse poses.

Qualitative Comparison.

Figure 4.6 provides a close-up view of facial regions across different fusion approaches. Each column corresponds to a specific method:

- **Reference:** Depicts the original subject or target pose for comparison.

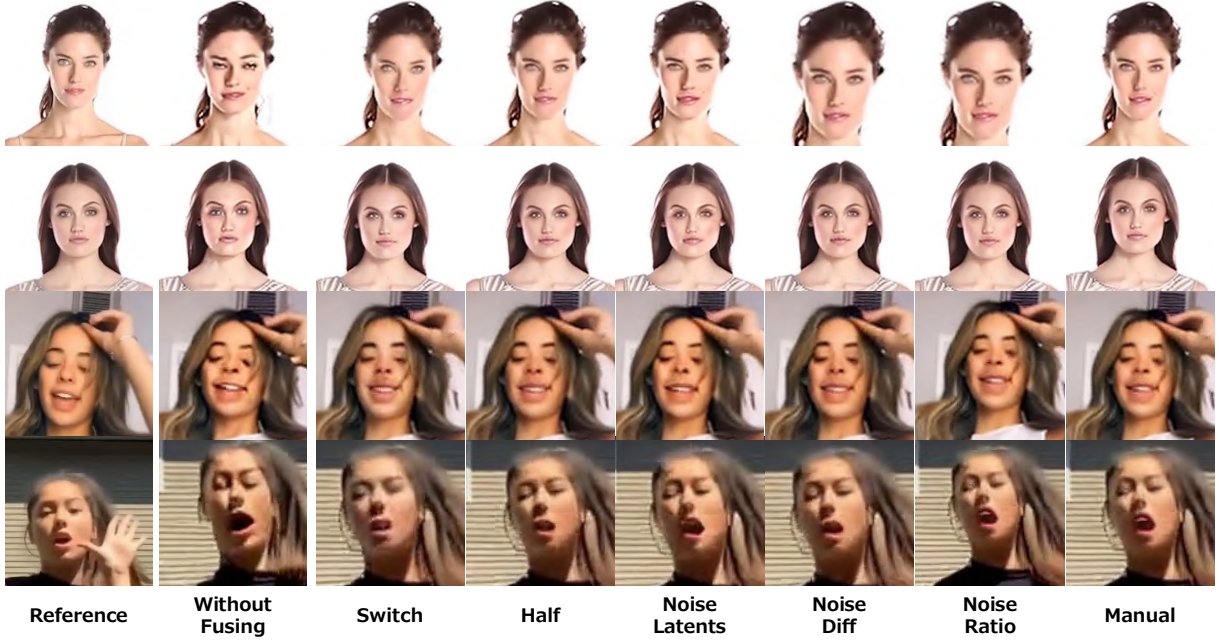


Figure 4.6 Qualitative ablation study. We compare the facial regions from videos generated by different fusion parameter settings and the baseline without fusion. Our proposed fusion methods consistently yield superior facial details, preserving subject identity and expressiveness. Notably, the ‘Manual’ and ‘Noise Latents’ settings demonstrate particular efficacy in generating realistic and visually appealing facial features, while the ‘Without Fusing’ baseline exhibits inconsistencies, and the ‘Switch’ and ‘Half’ methods show a tendency towards blur.

- **Without Fusing:** Uses a single network without any noise-level merging, often resulting in distorted or inconsistent faces.
- **Switch, Half, Noise Latents, Noise Diff, Noise Ratio:** Automated fusion schemes that blend noise residuals at each denoising step. “Switch” and “Half” occasionally suffer from blur or partial identity loss, whereas *Noise Latents*, *Noise Diff*, and *Noise Ratio* generally produce more coherent, stable faces.
- **Manual:** A human-in-the-loop strategy that fine-tunes the fusion parameter α_t based on visual inspection, often leading to the most realistic and expressive facial outcomes.

Visually, **Manual** and **Noise Latents** best preserve both facial detail and identity. In contrast, “Without Fusing,” “Switch,” and “Half” can exhibit more artifacts or blur, highlighting how noise-based fusion (with possible human oversight) is effective in generating clear, consistent facial features.

Figure 4.7 illustrates a similar set of fusion approaches but focuses on varying pose inputs. Columns represent distinct reference poses, while rows correspond to fusion methods (Without Fusing, Switch, Half, Noise Latents, Manual, etc.). We observe:

- **Without Fusing:** Produces inconsistent or artifact-laden facial renderings.
- **Switch, Half:** Relatively simple schedules that mitigate some distortions but risk blur or partial identity mismatch.
- **Noise Latents, Manual:** Offer sharper facial details and more stable expressions across frames, preserving identity more faithfully under different poses.

Hence, both *Noise Latents* and *Manual* emerge as standout strategies for balancing realistic appearance with strong subject fidelity.

4.5 Discussion

The proposed **AdaptiveFuse** framework demonstrates that noise-level fusion can effectively combine multiple specialized diffusion models, such as a motion-centric network (AnimateAnyone) and a face-refinement network (Face-Net), for dancing video generation. In doing so, it offers higher-quality outputs than previous methods in terms of both fidelity and consistency. Nevertheless, certain aspects of the approach warrant closer examination.

First, although the method excels at synthesizing clear and temporally coherent facial details in dancing videos, the computational overhead remains nontrivial. By employing multiple pre-trained models simultaneously, the system runs two specialized U-Nets in parallel, increasing the memory footprint and prolonging inference times. While our empirical results show the performance benefits of leveraging complementary modules, resource-constrained environments may require additional strategies such as parameter sharing, partial inference, or dynamic loading of specific model components.

Second, our exploration of the fusion parameter α_t has centered on either manual “human-in-the-loop” adjustments or heuristic noise-based schedules. Although these strategies are relatively easy to implement and validate, they may not adapt optimally to diverse motion scenarios



or complex facial expressions. Future work might incorporate more automated scheduling via reinforcement learning, gradient-based optimization, or adaptive heuristics that consider perceptual metrics and user-driven constraints. Such refinements could help fine-tune the relative weighting of motion and facial-detail models over time, leading to a more seamless blending of specialized capabilities.

Third, our empirical evaluation has focused on single-person dance sequences under controlled conditions. Scaling this approach to more challenging settings, such as multi-person choreography, highly cluttered scenes, or non-human entities, remains an open research avenue. For instance, modeling interactions among multiple dancers or handling occlusions in a dynamic group context may demand further architectural adaptations or enhanced fusion mechanisms. Likewise, tasks like object-centric video generation, full-scene stylization, and viewpoint transitions may also benefit from noise-level fusion, provided that future research validates the generality of this framework across broader tasks and data domains.

Finally, while our results highlight robust performance even under moderate complexities of motion and expression, situations involving extreme poses, rapid viewpoint changes, or severe lighting variations have not been exhaustively tested. In highly challenging scenarios, more advanced domain-specific priors or scene-level constraints may be necessary to preserve fidelity and ensure temporal coherence. Ongoing improvements to scheduling schemes, domain coverage, and model resiliency, alongside a deeper integration of real-time feedback signals, could strengthen the applicability of AdaptiveFuse and permit a wider range of creative and industrial uses.

Overall, the AdaptiveFuse approach offers a compelling demonstration of how noise-based model fusion can unify specialized strengths within a single pipeline, particularly for the demanding application of dancing video generation. By integrating a face-focused network with a motion-centric model, our strategy attains superior facial fidelity and temporal smoothness, pushing the envelope in identity-preserving human-video synthesis.



4.6 Summary

In this chapter, we extended the AdaptiveFuse framework to tackle the challenge of dancing video generation, focusing on two primary goals: capturing natural, fluid motion and preserving high-fidelity facial details. By fusing a motion-oriented model (AnimateAnyone-Net) with a face refinement module (Face-Net) at the noise-residual level, our approach achieves improved temporal coherence, more realistic facial features, and stronger identity consistency compared to existing methods.

A series of quantitative evaluations on benchmark datasets (TikTok and UBC) confirms that our method excels in preserving subject identity (Adaface Score) while maintaining video quality metrics (FVD, SSIM, PSNR) at or above the performance of baselines. Qualitative comparisons further underscore the effectiveness of this strategy: unlike conventional weight-centric or switch-based fusion, noise-level fusion enables each specialized model to contribute its strengths without mutually interfering. An ablation study demonstrates that both user-driven (Manual) and automated (Noise Latents) scheduling strategies enhance face fidelity, validating the flexibility and robustness of noise-based integration. These findings highlight the adaptability of AdaptiveFuse for demanding generative tasks in video domains and illustrate how noise-level fusion enables multiple specialized capabilities, motion accuracy and facial identity, such as to coexist within a single pipeline.

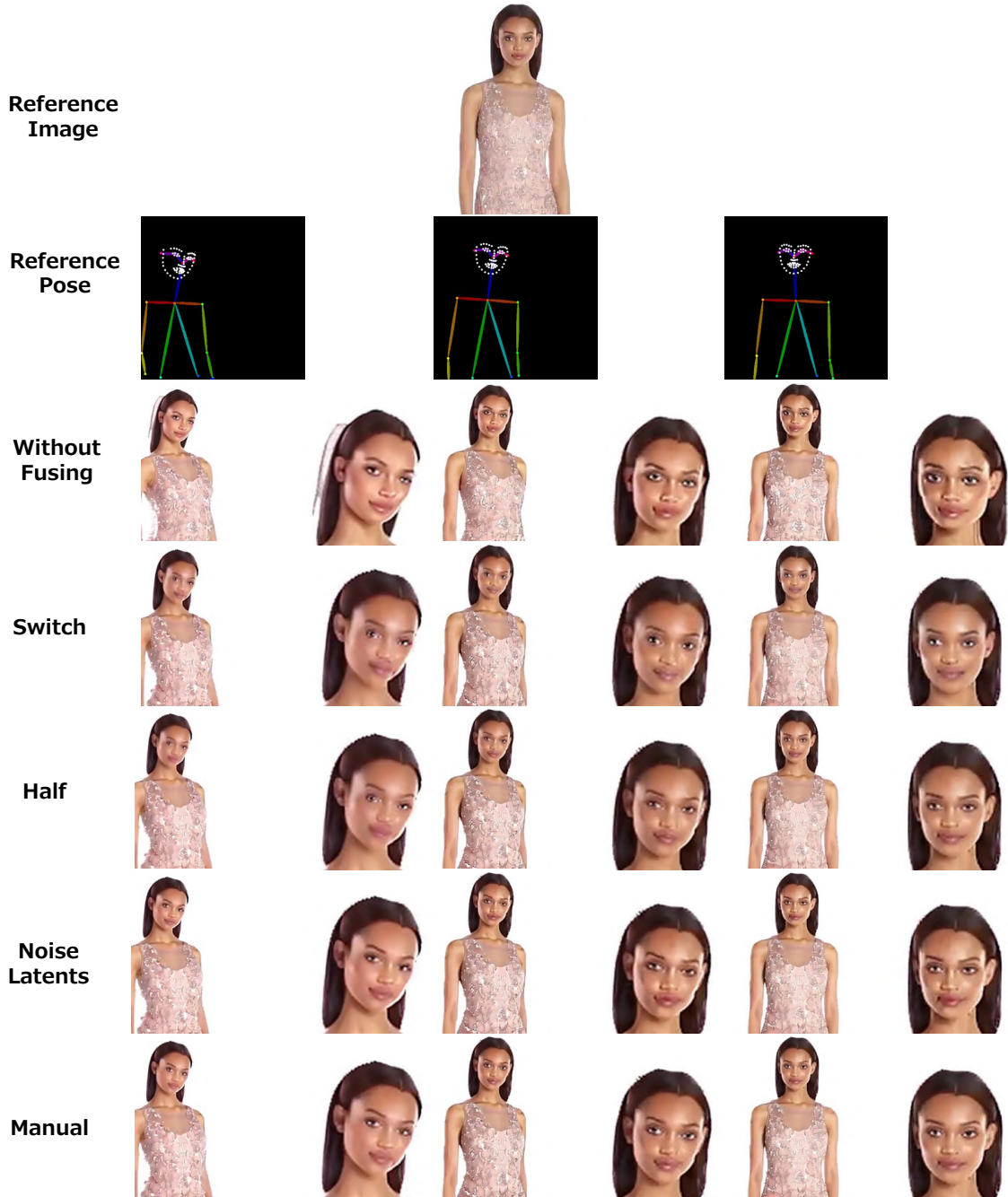


Figure 4.7 Ablation study visualizing the impact of different fusion parameter settings on facial region generation. We compare results from various fusion strategies against a baseline without fusion. Our proposed fusion methods consistently yield superior facial details, preserving subject identity and expressiveness. Notably, the 'Manual' and 'Noise Latents' settings demonstrate particular efficacy in generating realistic and visually appealing facial features, while the 'Without Fusing' baseline exhibits inconsistencies, and the 'Switch' and 'Half' methods show a tendency towards blur.





Chapter 5. CogVideoX with Multi-LoRA Fusion

In earlier chapters, we demonstrated how noise-based fusion can blend heterogeneous diffusion models or LoRA modules to better accommodate multiple objectives (e.g., style versus identity). However, real-world video generation tasks often require *several* specialized LoRAs simultaneously. For instance, one might need:

1. A **Camera Motion LoRA** to enforce cinematic viewpoint transitions (zoom, arc, push-in),
2. A **style LoRA** to implement a distinct aesthetic (e.g., cartoon shading, black-and-white film),
3. A **character LoRA** to anchor specific facial or bodily attributes of a known subject (identity preservation).

Merging all these aspects naively in a single pipeline can lead to parameter conflicts or partial overwrites. Additionally, many existing approaches only handle *one LoRA per pipeline*. This chapter addresses these constraints by extending our noise-based fusion (**AdaptiveFuse**) to handle multiple LoRAs across *two CogVideoX pipelines*, each focusing on a subset of tasks. We then fuse their noise predictions *step by step* during inference, enabling finer control of style, camera motion, and character identity.

5.1 Motivation and Research Questions

5.1.1 Motivation

Our motivation stems from the following challenges:

- **Multi-LoRA Interference:** If we load multiple LoRAs directly into a single pipeline (e.g., *camera + style + character*), we risk parameter collisions that can degrade certain LoRAs' effects.



- **Complex Task Requirements:** Users may demand vivid camera motions *and* stylized visuals *and* consistent facial features. Satisfying all three in one model often leads to compromised results.
- **Scalable Inference Without Re-Training:** We aim for a purely *inference-time* solution that avoids merging weights, so we can quickly switch or add new LoRAs if desired.

By dedicating **Transformer model A** to Camera Motion LoRA and character LoRAs and **Transformer model B** to style LoRAs, we hypothesize that noise-level fusion will exploit each pipeline’s specialized knowledge while circumventing destructive interference.

5.1.2 Research Questions

- **RQ1:** Does splitting multiple LoRAs into two pipelines (and fusing their noise predictions) yield higher *text alignment* and *style fidelity* than naive single-pipeline merges?
- **RQ2:** Can we preserve **character identity** and **camera motion** effectively when also applying a strong style LoRA?
- **RQ3:** How do different scheduling methods (*stable, linear, exponential, manual*) for α_t affect final video consistency and quality?

We address these questions using CogVideoX, which is a powerful text-to-video framework, as our base model, plus multiple LoRAs specialized for camera, style, and identity.

5.2 Methodology

5.2.1 Two-model Setup in CogVideoX

We use two instances of CogVideoX (dubbed **Transformer model A** and **Transformer model B**), each referencing the same base weights but loading different LoRAs:

Transformer model A

- *Character LoRA*: A rank-constrained adapter that focuses on preserving a specific face or body shape (e.g., a cartoon hero, a real-world celebrity).
- *Camera Motion LoRA*: Another LoRA that encodes specialized camera motions (zoom, dolly, arc) and framing preferences.

Transformer model B

- *Style LoRA*: A rank-constrained adapter that imposes a particular aesthetic (e.g., Disney, black-and-white anime, watercolor).

During inference, each pipeline maintains its own 3D transformer and autoencoder modules for the video latents, although they share the same sampling schedule. We keep them on *CPU* when idle and transfer them to *GPU* only when computing noise predictions, mitigating memory overhead.

Figure 5.1 illustrates the algorithmic pipeline of our approach. The process consists of three main components: two separate CogVideoX transformers and a fusion module that fuses their outputs.

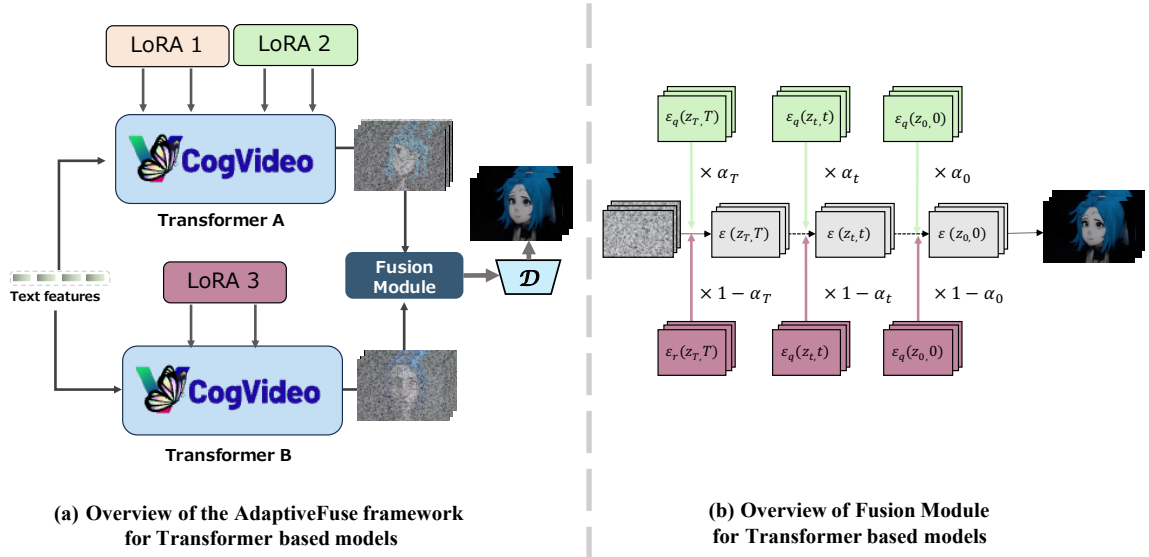


Figure 5.1 Overview of the Multi-LoRA Fusion Framework. (a) Two CogVideoX transformer networks, each loaded with different LoRA sets, receive the same text features but generate distinct noise predictions. (b) The Fusion Module dynamically combines these predictions across denoising steps using scheduled α weights to produce the final video.

5.2.2 Noise-Level Fusion

At each diffusion timestep t , let:

- $\epsilon_A(\mathbf{z}_t, t)$ is the noise residual output from Transformer model A.
- $\epsilon_B(\mathbf{z}_t, t)$ is the noise residual output from Transformer model B.

We combine these two noise predictions as:

$$\epsilon(\mathbf{z}_t, t) = \alpha_t \times \epsilon_A(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_B(\mathbf{z}_t, t), \quad (5-1)$$

where the scalar weight α_t can evolve according to a specified *schedule* (e.g., *linear*, *power*, *cosine*, or *step*). We also allow a *reversed* blend by letting $\alpha_t \mapsto 1 - \alpha_t$ if we want to favor style LoRA more in early timesteps.

Finally, we pass $\epsilon_{\text{fused},t}$ to the scheduler (DDIM, DPM, etc.) to obtain the latents $\mathbf{z}_{t-1}$. This iterative approach ensures that camera + character refinements from $\epsilon_{A,t}$ and style refinements from $\epsilon_{B,t}$ co-occur at the *noise* level, rather than being forced into a single weight space.

5.2.3 Alpha Schedules for Multi-LoRA Fusion

Recall from Eq. (5-1) that we blend the two pipelines' noise predictions as

$$\epsilon(\mathbf{z}_t, t) = \alpha_t \times \epsilon_A(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_B(\mathbf{z}_t, t), \quad (5-2)$$

where t runs from the final diffusion step T down to 1 and $\alpha_t \in [0, 1]$ balances how strongly Transformer model A (camera + character) versus Transformer model B (style) influences the final noise. The parameter α_t itself can vary over time according to different *scheduling functions*. Below, we formalize three types of schedules.

(1) Linear Schedule. Let α_{start} and α_{end} be constants in $[0, 1]$. We define a **linear** schedule by:

$$\alpha_t = \alpha_{\text{start}} + (\alpha_{\text{end}} - \alpha_{\text{start}}) \cdot \frac{t}{T}, \quad (5-3)$$

where $t \in \{0, 1, \dots, T\}$ indexes the denoising steps in ascending order (for convenience). For example, if $\alpha_{\text{start}} = 0.3$ and $\alpha_{\text{end}} = 0.9$, then $\alpha_0 = 0.3$ and $\alpha_T = 0.9$, and α_t increases

uniformly from 0.3 to 0.9 as t goes from 0 to T . This schedule smoothly transitions from favoring Transformer model B (if $\alpha_{\text{start}} < 0.5$) to favoring Transformer model A (if $\alpha_{\text{end}} > 0.5$), or vice versa, depending on the chosen values. In this thesis, we set two linear schedules:

- $\alpha_{\text{start}} = 0.2$ and $\alpha_{\text{end}} = 0.8$
- $\alpha_{\text{start}} = 0.4$ and $\alpha_{\text{end}} = 1.0$

(2) Step (Piecewise-Constant) Schedule. A **step** schedule is useful for abruptly shifting α_t at certain fractions of T . Suppose we define a sequence of breakpoints $0 = s_0 < s_1 < \dots < s_k < s_{k+1} = T$ and corresponding α -levels $a_1, a_2, \dots, a_{k+1}$ in $[0, 1]$. The piecewise-constant function can be written as:

$$\alpha_t = \begin{cases} a_1, & \text{if } 0 \leq t < s_1, \\ a_2, & \text{if } s_1 \leq t < s_2, \\ \vdots & \\ a_{k+1}, & \text{if } s_k \leq t \leq T. \end{cases} \quad (5-4)$$

For instance, one might choose three jumps:

$$(s_0, s_1, s_2, s_3) = (0, \frac{T}{3}, \frac{2T}{3}, T), \quad (a_1, a_2, a_3) = (0.3, 0.6, 0.8, 1.0),$$

causing α_t to jump from 0.3 to 0.6 at $t = \frac{T}{3}$ and from 0.6 to 0.8 at $t = \frac{2T}{3}$, and finally 1.0 near the end. This can produce an abrupt shift in style vs. camera weighting, which may or may not be desirable depending on the application.

Figure 5.2 illustrates five different step-based scheduling patterns. The step1 schedule (blue) implements a classic 5-stage progression with equal intervals, gradually increasing from 0.2 to 1.0. The step2 schedule (orange) follows a pattern similar to our example above, with an early jump to 0.6 at $t = 0.1 \times T$. The step3 schedule (green) is back-loaded, maintaining low alpha values until later stages, thus preserving Model B’s influence for most of the diffusion process. The step4 schedule (red) applies a gradual increase with accelerated jumps in the middle range. The step5 schedule (purple) introduces a non-monotonic pattern with a temporary decrease from 0.7 to 0.5 in the middle stages, creating a unique wave-like progression that

can produce interesting blending effects. This variety of step functions provides flexible control over how the two models interact throughout the diffusion process, allowing for creative experimentation with different transition patterns.

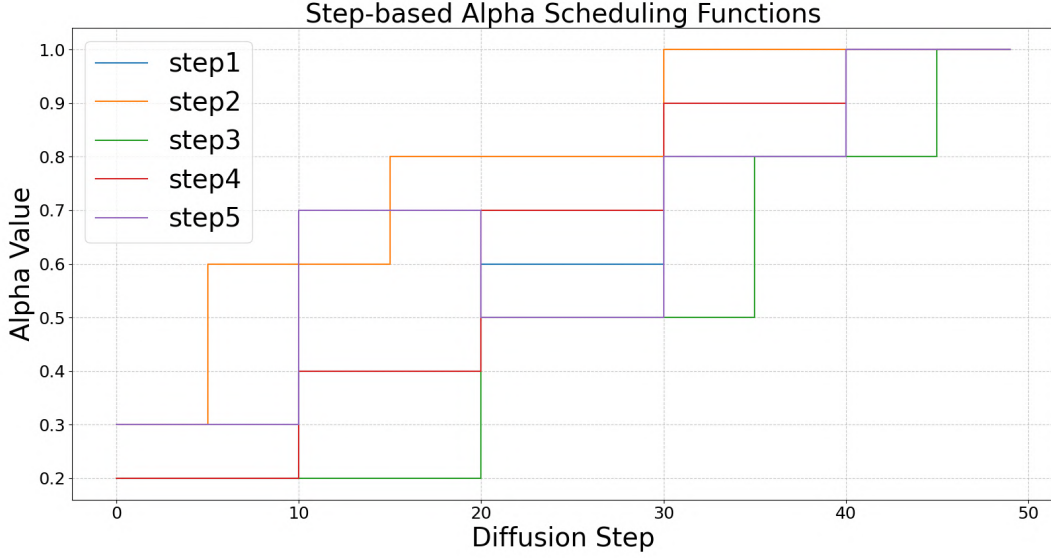


Figure 5.2 Step-based Alpha Scheduling Functions for CogVideoX multi-LoRA fusion. The graph shows five different piecewise-constant scheduling patterns that control the blending weights between two transformer models throughout the diffusion process. Step1 (blue) implements a uniform 5-stage progression; Step2 (orange) features early jumps favoring Model A sooner; Step3 (green) is back-loaded to preserve Model B’s influence longer; Step4 (red) applies a gradual increase with accelerated midpoint progression; and Step5 (purple) introduces a unique non-monotonic pattern with a temporary decrease in the middle stages, creating a wave-like blending effect.

(3) Power (Polynomial) Schedule. A power schedule (sometimes called “polynomial”) can speed up or slow down the rate of change. Let $\gamma > 0$ be a power exponent. Then:

$$\alpha_t = \alpha_{\text{start}} + (\alpha_{\text{end}} - \alpha_{\text{start}}) \left(\frac{t}{T} \right)^\gamma. \quad (5-5)$$

If $\gamma > 1$, α_t stays near α_{start} longer, then rapidly approaches α_{end} in later steps. If $0 < \gamma < 1$, we get the opposite: a fast initial rise that slows near the end. This provides a more flexible “curved” schedule than the linear function in Eq. (5-3). In this thesis, we set two power schedules:

- $\alpha_{\text{start}} = 0.2$, $\alpha_{\text{end}} = 0.8$ and $\gamma = 3$
- $\alpha_{\text{start}} = 0.4$, $\alpha_{\text{end}} = 1.0$ and $\gamma = 3$



Practical Implications. These schedules allow the user to tailor α for specific tasks—for example, “lock in identity early, add heavy stylization near the end” can be handled by a standard linear schedule. Conversely, “emphasize style right away” uses reversed blending. This flexibility is crucial for multi-LoRA scenarios where the user has competing objectives (detailed camera moves, strong stylization, identity fidelity) that need to be balanced over the denoising process.

Overall, the shape of α_t strongly influences how the final frames evolve, since an LoRA that is weighted more heavily in early steps tends to define the broad scene composition while LoRAs emphasized later can refine details such as facial identity or subtle stylistic cues.

5.2.4 Algorithmic Overview

Algorithm 3 formalizes this procedure.

Algorithm 3 CogVideoX Multi-LoRA Noise Fusion

Require: Text prompt p , Number of steps T , LoRAs (1, 2, 3), α schedule

Ensure: Generated video frames

- 1: Initialize Transformer model A with LoRA 1 + LoRA 2
 - 2: Initialize Transformer model B with LoRA 3
 - 3: Sample initial noise latent z_T
 - 4: **for** $t = T$ **down to** 1 **do**
 - 5: $\epsilon_A(\mathbf{z}_t, t) \leftarrow$ Transformer model A(z_t, p, t)
 - 6: $\epsilon_B(\mathbf{z}_t, t) \leftarrow$ Transformer model B(z_t, p, t)
 - 7: Calculate α_t based on selected schedule
 - 8: $\epsilon(\mathbf{z}_t, t) \leftarrow \alpha_t \times \epsilon_A(\mathbf{z}_t, t) + (1 - \alpha_t) \times \epsilon_B(\mathbf{z}_t, t)$
 - 9: $z_{t-1} \leftarrow \epsilon(\mathbf{z}_t, t)$
 - 10: **end for**
 - 11: Decode z_0 to pixel space using VAE decoder
 - 12: **return** Video frames
-

5.3 Experimental Setup

5.3.1 Base Models

Our experiments use CogVideoX [29] as the foundational text-to-video diffusion model. CogVideoX represents a significant advancement in video generation, combining a 3D transformer architecture with diffusion-based denoising. The model operates primarily in a compressed latent space to reduce computational requirements, similar to the approach used in Stable Diffusion for images.

Key characteristics of the CogVideoX model include:

- **Architecture:** A 3D U-Net transformer optimized for spatio-temporal consistency
- **Parameters:** Approximately 5 billion parameters in the base model
- **Latent Representation:** Uses a VAE to compress video frames into a lower-dimensional latent space
- **Text Conditioning:** Employs cross-attention mechanisms to integrate text guidance from a T5 text encoder
- **Temporal Modeling:** Includes dedicated temporal attention layers to maintain consistency across frames

The model has demonstrated strong capabilities in generating coherent videos from textual descriptions, making it an ideal candidate for our multi-LoRA fusion experiments. Importantly, its architecture supports efficient adaptation through LoRA modules, allowing us to enhance specific attributes (camera motion, style, character features) without full model retraining.

5.3.2 LoRAs and Prompts

We conduct experiments on short, 24-frame videos generated at 480×720 resolution. We test three public LoRAs:

- *CineCam LoRA (camera motions)* [102],



- *Arcane-Character LoRA (character) [103],*
- *Disney-Style LoRA (alternative style target) [104],*

We combine these LoRAs in different ways across Pipeline A and Transformer model B to highlight synergy vs. conflict. Prompts typically describe camera movement ("Camera Arc", "Camera Pushin"), a certain style ("BW_STYLE", "Disney-like lighting"), a character ("Arcane"), and textual guidance about the subject ("a young woman with blue hair in a neon city"). We prepare for 10 prompts and set 3 seeds, and generate 30 videos in each setting.

5.3.3 Implementation Details

Our implementation is built on the CogVideoX framework, a powerful text-to-video model that supports LoRA fine-tuning. All experiments were conducted using a single NVIDIA RTX 3090 GPU with 24GB of VRAM.

To optimize memory usage during inference (particularly important for video generation), we keep only one transformer on GPU at a time, swapping models in and out as needed. This strategy allows us to process longer video sequences even with limited GPU memory, as the two transformers never need to be simultaneously present in VRAM during the forward pass.

5.3.3.1 Model Configuration

We use CogVideoX 5B as our base model, which is loaded twice (once for each Transformer model) with separate LoRA modules. The model configuration follows:

- **Resolution:** 480×720 pixels
- **Video Length:** 24 frames (8 FPS)
- **Number of Inference Steps:** 50 steps using a DDIM scheduler
- **Classifier-Free Guidance Scale:** 6.0



5.3.3.2 LoRA Configuration

Following the approach described in Chapter 3, we carefully configure our LoRA modules:

- **Transformer model A:** Loads CineCam LoRA (weight: 0.5) and Arcane-Character LoRA (weight: 0.5)
- **Transformer model B:** Loads Disney-Style LoRA (weight: 1.0)

The merge weights within each Transformer model are controlled through the `set_adapters` function, as shown in the code:

5.3.4 Baselines

We compare the following approaches:

1. **No LoRA:** CogVideoX with only text conditioning, no LoRA modules.
2. **All-in-One Merge:** Attempting to load all LoRAs (camera + style + character) into a single pipeline, merging them at the weight level.
3. **Switch:** Alternating LoRAs at fixed intervals in a single Transformer model, without residual blending.
4. **Ours (Multi-LoRA Fusion):** Two Transformer models, each with specialized LoRAs, combined via noise-level fusion in Eq. (5-1).

5.3.5 Evaluation Metrics

We quantify performance with:

- **Text CLIP Score** (higher is better): Measures alignment between generated frames and the textual prompt.
- **Temporal Consistency** (closer to 1.0 is better): Assesses frame-to-frame coherence using semantic consistency between consecutive frames. Following EvalCrafter, we compute

the cosine similarity between CLIP embeddings of adjacent frames (CLIP-Temp) and average these similarities across the video [101].

- **Face Consistency** (closer to 1.0 is better): Evaluates stability of facial identity across frames. We use the first frame as reference and calculate the average cosine similarity between face embeddings of the first frame and all subsequent frames, using EvalCrafter [101].

5.4 Results and Discussion

5.4.1 Quantitative Results

Table 5.1 reports the averaged scores for four different methods across multiple test prompts:

Table 5.1 Quantitative comparisons of multi-LoRA usage in CogVideoX.

Method	Text CLIP $\uparrow$	Temp. Consistency $\uparrow$	Face Consistency $\uparrow$
No LoRA	32.96	0.9956	0.9830
Weight Merging	34.88	0.9959	0.9824
Switch	35.09	0.9963	0.9779
Ours	35.61	0.9970	0.9834

We make several observations:

- **Text CLIP Score:** Our fusion obtains the highest text alignment (35.61), surpassing the Switch method (35.09), Weight Merging (34.88), and No LoRA (32.96). By separating camera/character from style at the noise level, the system more faithfully realizes prompt details.
- **Temporal Consistency:** The highest temporal consistency is also achieved by our approach (0.9970). Even though Switch obtains 0.9963, the difference indicates fewer flickers or artifacts in the fused Transformer model’s output.

- **Face Consistency:** Our method yields the best face consistency (0.9834), indicating stronger identity preservation across frames than other baselines. Notably, Switch obtains 0.9779, and Weight Merging yields 0.9824, both below ours.

These results demonstrate that multi-LoRA noise-fusion not only balances style and camera motion, but also maintains subject fidelity, outperforming simpler merges or Transformer model switching.

5.4.2 Qualitative Results

Figure 5.3 illustrates video frames at selected frames for four different approaches applied to the same text prompt. Each row represents a distinct method for generating a scene of a blue-haired woman overlooking a cityscape with a dolly zoom camera effect.

Text Prompt: CSETIARCANE. Dolly Zoom. BW_STYLE. High above a neon-lit cityscape, a young woman with vivid blue hair leans against a rusted railing. A black and white animated scene. As the camera performs a slow dolly zoom toward her, the distant city lights blur and warp behind her silhouette. Her expression transitions from restless anticipation to a calm, determined gaze, the faint hum of arcane energy palpable in the night air.



Figure 5.3 Comparison of video generation methods using text prompt. Each row shows frames from videos generated by different approaches: No LoRA (baseline without specialized modules), Weight Merging (all LoRAs merged at parameter level), Switch (alternating between LoRAs at specific steps), and Ours (noise-based fusion of specialized Transformer models). Note how our method maintains both consistent character identity and appropriate camera motion while incorporating the black and white stylization.

- **No LoRA** (top row) produces basic content that follows the prompt but lacks distinctive stylization or character features. The camera motion works well, but the facial details

appear generic with inconsistent expressions across frames.

- **Weight Merging** (second row) shows improved character identity with blue hair, but still struggles with the character identity generation. While some stylistic elements emerge, but there are still have some colors even we set the black and white style setting..
- **Switch** (third row) produces visually distinct frames with stronger anime stylization. The background keeps in black and white style, but the zoom is subtle. Also, the character appearance varies from original target.
- **Ours** (bottom row) demonstrates superior balance of all required elements: consistent character identity with the recognizable Arcane style, proper camera movement with gradual perspective changes that create the dolly zoom effect, and cohesive background elements that blur appropriately as the camera moves in. Facial expressions remain stable while subtly transitioning as specified in the prompt.

The comparison highlights how our multi-LoRA fusion approach effectively integrates camera motion, style, and character identity in a single coherent generation. Unlike other methods that either sacrifice one aspect for another or produce inconsistent results, our noise-based fusion preserves the specialized capabilities of each LoRA without destructive interference. This is particularly evident in the consistent facial features combined with the cinematically appropriate background perspective changes.

5.4.3 Ablation Studies

To understand how different alpha schedules impact the generation quality, we conducted extensive experiments with various scheduling strategies. Figure 5.4 illustrates the impact of different alpha scheduling strategies on video generation quality. The rows represent different scheduling methods applied to the same prompt and seed. The fixed weight approaches ($\alpha = 0, \beta = 1$ and $\alpha = 1, \beta = 0$) show clear trade-offs: the first emphasizes style but loses character details, while the second prioritizes character but produces less distinctive styles.



The power-based schedules with $\alpha_{\text{start}} = 0.2$ and $\alpha_{\text{start}} = 0.4$ demonstrate how higher starting values help preserve character identity while still capturing stylistic elements. Step-based schedules (step2 and step5) create more defined transitions between style and character emphasis, sometimes leading to more pronounced scene composition. Their reversed counterparts show alternative progression patterns that emphasize different aspects of the generation.

Text Prompt: CSETIARCANE. Camera Tracking. BW_STYLE. A young woman with vivid blue hair examines a wall of glowing hexagonal displays in an underground control center. A black and white animated scene. The room is bathed in soft, pulsing light from the screens, creating a pattern of highlights across her concentrated features. As the camera tracks slowly sideways, revealing more of the curved wall of technology, her expression gradually shifts from analytical focus to dawning comprehension. The ceiling is low and industrial, with pipes and cables forming complex patterns above.

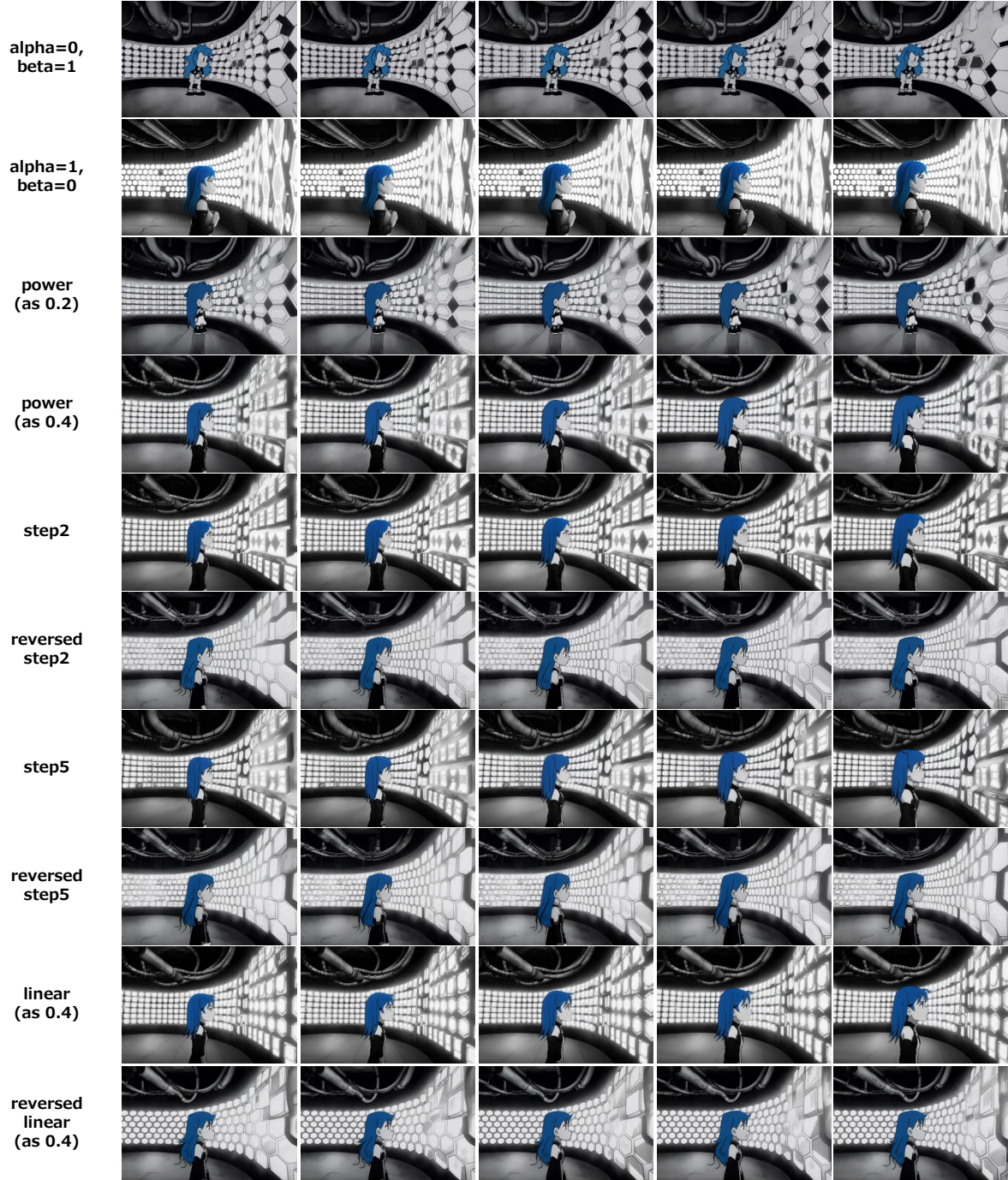


Figure 5.4 Comparison of different alpha scheduling strategies for multi-LoRA fusion. The figure shows frame sequences from videos generated using various scheduling methods: fixed weights (top two rows), power-based schedules with different starting values, step-based schedules (step2 and step5) in both standard and reversed configurations, and linear schedules. Each row represents a different $\alpha(t)$ schedule applied to the same text prompt describing a young woman examining displays in a control center. Note how different schedules affect camera movement quality, character fidelity, and environmental details. The linear schedule with $\alpha_{\text{start}} = 0.4$ (second row from bottom) achieves the highest Text CLIP score (35.61), balancing style elements with consistent character identity.



Linear schedules, particularly with $\alpha_{\text{start}} = 0.4$, achieve the most balanced results, maintaining both the hexagonal display details from the style LoRA and the consistent character features from the character LoRA. This balance is reflected in the quantitative metrics, where this configuration achieves the highest Text CLIP score of 35.61.

Our analysis in Table 5.2 shows that linear schedules with $\alpha_{\text{start}} = 0.4$ and power schedules ($\alpha_{\text{start}} = 0.4$) both achieve the highest Text CLIP scores of 35.61, indicating superior prompt alignment. Notably, balanced fixed weights ($\alpha = 0.5, \beta = 0.5$) also perform well at 35.46, suggesting both Transformer models contribute valuable information throughout the denoising process.

Based on both quantitative results and visual quality assessment:

- **Step-based schedules** with abrupt transitions often produce visible style shifts mid-generation. They can achieve good Text CLIP scores (up to 35.43 for step5).
- **Power-based schedules** allow Transformer model A (camera+character) influence to remain low initially before rapidly increasing later in the denoising process. This approach keeps α_t near α_{start} for early steps, then accelerates toward α_{end} . Our experiments with $\alpha_{\text{start}} = 0.4$ yield exceptional results (35.61), creating smooth transitions between style and character emphasis.
- **Linear schedules** with moderate α_{start} values (0.4) provide a more balanced approach, gradually increasing Transformer model A’s influence. This produces videos with well-established stylistic elements while maintaining consistent character identity and camera motion, achieving our highest overall performance (35.61).
- **Reversed schedules**, which favor Transformer model A initially and then increase Transformer model B’s influence, sometimes outperform their standard counterparts. For instance, reversed linear with $\alpha_{\text{start}} = 0.2$ scores 35.45, compared to 35.33 for the standard version. This suggests that establishing character and camera attributes early, then refining with style later, can be effective for certain prompts.



Manual tuning based on visual inspection remains valuable for specialized content. When fine-tuning schedules for individual prompts, we found that emphasizing camera and character early (to establish composition and identity) followed by increased style influence later (for aesthetic refinement) generally produces the most coherent results. However, the optimal schedule varies depending on the desired balance between camera dynamics, character fidelity, and stylistic intensity.

5.5 Discussion

5.5.1 Diffusion Process and Latent Space Structure

The theoretical foundation of our multi-LoRA fusion approach is rooted in the unique properties of diffusion models' latent manifolds. In diffusion-based generative models, the denoising process gradually transforms random noise into structured data by reversing a forward diffusion process. This iterative transformation creates a pathway through the latent space that differs fundamentally from parameter-space representations. When a diffusion model is fine-tuned with a LoRA, the adaptation primarily affects how the model navigates this denoising trajectory. Each specialized LoRA effectively creates a biased pathway through the latent space, emphasizing certain features (camera motions, stylistic elements, character attributes) at different stages of the denoising process. Unlike GAN-based approaches where interpolation often leads to unrealistic artifacts, diffusion models offer smoother transitions between different generative objectives.

5.5.2 Information Preservation in Noise Predictions

A key theoretical insight behind our approach is that noise predictions from different LoRA-adapted models contain complementary information that can be meaningfully combined. The noise residual $\epsilon_\theta(z_t, t)$ at each timestep encapsulates the model's learned prior about how to incrementally transform the current latent toward the data distribution. When models specialize in different aspects of generation, their noise residuals focus on distinct aspects of this transformation. By blending these noise predictions, rather than model parameters, we preserve each

model’s specialized knowledge without the destructive interference that occurs in weight-space merging.

5.5.3 Theoretical Basis for Alpha Scheduling

The effectiveness of different α_t schedules can be understood through the lens of how information is encoded across diffusion timesteps. At early denoising steps (high noise levels), the model primarily determines broad structural elements and composition. As denoising progresses, finer details emerge. This natural progression suggests that different LoRAs may have ”optimal influence periods” during the denoising process.

For example, camera motion LoRAs may be most effective at establishing the overall scene geometry in earlier steps, while style LoRAs might better refine texture and lighting details in later steps. This theoretical understanding explains why our experiments found that linear and power-based schedules with moderate α_{start} values often outperform fixed weights or abrupt step schedules: they align better with the natural information encoding across the diffusion trajectory.

5.5.4 Multiple Objective Optimization Without Retraining

From an optimization perspective, our approach represents a form of non-parametric ensemble learning applied to generative trajectories. Traditional multi-objective optimization in neural networks often requires explicit joint training with carefully weighted loss functions. In contrast, our noise-fusion approach allows multiple objectives to be balanced, without re-opening the complex optimization problem. This can be viewed theoretically as a form of Pareto-optimal balancing between competing objectives (camera quality, style fidelity, character preservation) in the space of generative trajectories rather than in parameter space. The dynamic α_t scheduling allows users to find an optimal balance for specific generation tasks without retraining the underlying models.



5.5.5 Theoretical Limitations

It’s important to note the theoretical limitations of noise-based fusion. The assumption that noise predictions can be linearly combined presupposes that the latent space structure is relatively consistent between different LoRA adaptations of the same base model. If LoRAs substantially alter the semantic meaning of latent dimensions or create highly divergent generative trajectories, linear combination may produce unrealistic results or collapse to averaging artifacts. Additionally, our approach assumes temporal and spatial consistency in how different LoRAs encode information across the diffusion process. If LoRAs encode similar information at drastically different denoising timesteps, even optimal α_t scheduling may fail to properly reconcile these differences without more sophisticated temporal alignment techniques.

5.6 Summary

This chapter presented an extension of our *AdaptiveFuse* noise-based paradigm to handle *multiple LoRAs* across two CogVideoX Transformer models. By allowing each Transformer model to focus on a subset of LoRAs (camera + character vs. style), then fusing their predicted noise residuals at each denoising step, we achieve:

- Stronger text alignment with specified camera moves,
- Distinct, high-quality stylization,
- Sustained subject identity, even in motion scenes,

surpassing single-Transformer model fuses in both objective metrics and subjective comparisons. While memory overhead and scheduling complexity remain concerns, the flexibility gained for multi-objective video generation is significant.

Table 5.2 Ablation study on alpha scheduling methods with multiple metrics.

Schedule Type	Text CLIP $\uparrow$	Temporal Consistency $\uparrow$	Face Consistency $\uparrow$
Fixed Weights			
$\alpha = 0, \beta = 1$	34.07	0.9945	0.9723
$\alpha = 0.5, \beta = 0.5$	35.46	0.9966	0.9843
$\alpha = 1, \beta = 0$	34.24	0.9979	0.9900
Linear			
linear ($\alpha_{\text{start}} = 0.2$)	35.33	0.9954	0.9779
linear ($\alpha_{\text{start}} = 0.4$)	35.61	0.9970	0.9834
rev. linear ($\alpha_{\text{start}} = 0.2$)	35.45	0.9971	0.9851
rev. linear ($\alpha_{\text{start}} = 0.4$)	35.18	0.9967	0.9858
Power-based			
power ($\alpha_{\text{start}} = 0.2$)	34.78	0.9948	0.9724
power ($\alpha_{\text{start}} = 0.4$)	35.61	0.9965	0.9826
rev. power ($\alpha_{\text{start}} = 0.2$)	35.21	0.9973	0.9870
rev. power ($\alpha_{\text{start}} = 0.4$)	35.31	0.9969	0.9851
Step-based			
step1	35.07	0.9952	0.9761
step2	35.27	0.9965	0.9812
step3	34.85	0.9951	0.9740
step4	35.05	0.9950	0.9765
step5	35.43	0.9964	0.9794
rev. step1	34.94	0.9975	0.9854
rev. step2	35.08	0.9967	0.9863
rev. step3	34.95	0.9977	0.9875
rev. step4	34.89	0.9973	0.9864
rev. step5	35.36	0.9970	0.9830



Chapter 6. Conclusion and Future Work

6.1 Conclusion

Generative modeling with diffusion-based methods has swiftly progressed, opening up new frontiers in image stylization, personalized content creation, and video generation. Yet, the real-world need for *multi-objective* capabilities, such as blending distinctive styles, preserving specific subject identities, and synthesizing complex motion, has proven challenging for a single diffusion model. To address this gap, we present AdaptiveFuse, a *noise-based fusion framework* that merges specialized diffusion models at inference time without re-training, regardless of differences in architecture or sampling schedules. As shown in Chapter 3, we demonstrated style-personalized images, and in Chapter 4 we extended that to dancing videos by fusing AnimateAnyone-Net with Face-Net. Chapter 5 further expanded the framework to CogVideoX, enabling the fusion of multiple LoRAs across separate transformer pipelines to simultaneously control camera motion, character identity, and stylistic elements in generated videos.

The core insight lies in manipulating *noise residuals* generated by individual models, rather than merging their weights or switching entire modules. By combining these residuals dynamically at each denoising step, each specialized model, such as a face-refinement module or a style LoRA, retains its domain-specific strengths. At the same time, the unified system generates coherent outputs that balance diverse requirements: high-fidelity facial features, targeted stylistic elements, and robust motion consistency. Experiments on image-based tasks showed that this noise-fusion approach better preserves both style and subject identity than simpler weight-based merges. Extending the same principle to video domains, particularly dancing video generation, demonstrated tangible improvements in facial fidelity, temporal smoothness, and overall video quality.

Additionally, analyses revealed the flexibility afforded by noise-based scheduling. Manual tuning or heuristic-driven schedules enable the fusion process to adapt as each frame is denoised,



giving finer control over which model’s outputs dominate at specific timesteps. This step-by-step balancing proved particularly effective for handling motion-rich sequences, where global dynamics and local face details often conflict. Despite these successes, running multiple models concurrently entails higher computational costs and requires intelligent scheduling to avoid artifacts or excessive blur.

Practical Value and Computational Efficiency

While the above results underscore the *technical* efficacy of our noise-fusion approach, its practical value also becomes evident when viewed from an application perspective. The key benefit is that users can seamlessly combine multiple specialized models, such as a character LoRA and a style LoRA, *without retraining*. In content creation workflows (animation, game development, or marketing media), this capability drastically lowers the production overhead:

- **Faster Iteration and Reusability:** Once trained, each LoRA module (a style module, an identity module, a motion module, etc.) can be reused and mixed at inference time, eliminating the need for repeated training cycles. This accelerates creative decision-making, as users can quickly test multiple visual variants within minutes.
- **Computation Cost Savings:** LoRA retraining can consume hours or days of GPU time. In contrast, our method only requires inference computations, enabling new combinations to emerge in a fraction of that time. This difference is crucial for resource-constrained users or smaller studios that lack large-scale server infrastructure.
- **Accessibility for Broader User Groups:** Since our approach requires no additional fine-tuning, individuals or organizations with minimal compute resources can still benefit from specialized LoRAs (publicly available or third-party). The adoption barriers for advanced generative workflows are thus lowered, making creative AI more inclusive.

These points illustrate how **AdaptiveFuse** transcends pure research demonstrations to address real-world production constraints. It offers a plug-and-play mechanism that can be readily



extended to 2D image editing, 3D scene generation, or multi-character storytelling, all while mitigating training overhead.

Overall, we underscore that *noise-based fusion* can be a practical and versatile solution for integrating heterogeneous diffusion models into a single generative pipeline. By abstracting away from weight-level constraints, it unlocks broader creativity and functionality, ranging from stylized portraiture to identity-consistent dancing videos, without sacrificing the stability and robustness that diffusion methods offer.

6.2 Future Work

Although AdaptiveFuse successfully addresses multi-objective generation across diverse tasks, several directions remain open for further exploration. In particular, the discussions in earlier chapters suggest that advancing efficiency, refining scheduling, expanding application domains, and improving robustness under challenging conditions can all elevate the impact of noise-fusion techniques. Below, we summarize four key directions for future investigations:

1. Efficiency and Memory Constraints. While combining multiple specialized U-Nets broadens the functional scope of the pipeline, it also increases memory usage and inference time. Pursuing partial-inference strategies or model compression may help mitigate the added resource demands, enabling wider deployment across devices with limited computational capacity.

2. Enhanced Scheduling Algorithms. Current scheduling methods rely on heuristics or manual interventions for deciding how to fuse noise residuals. A more data-driven or adaptive approach, such as reinforcement learning or gradient-based optimization, could automatically balance the influence of sub-models over time. Such an automated scheduler would not only streamline practical usage but also potentially yield more consistent results in dynamic scenarios, such as human motion.

3. Broader Application Domains. Although this thesis highlighted dancing video generation and stylized image synthesis, the same noise-fusion paradigm may benefit tasks like multi-person interactions, complex scene modifications, and novel view synthesis. Investigat-



ing how AdaptiveFuse performs on challenging domains, such as large-scale 3D environments or cross-modal generation, represents a natural next step in expanding its utility.

4. Robustness to Challenging Conditions. Future work should further test AdaptiveFuse under conditions involving occlusions, style variations, abrupt body motion, and multi-person scenes. Introducing domain-specific models or integrative scene models could bolster the framework’s resilience to unpredictable environments, thereby fostering more reliable real-world deployments.

By pursuing these directions, subsequent research can amplify AdaptiveFuse’s flexibility, computational efficiency, and overall robustness, ensuring that noise-based fusion stands as a robust and widely applicable technique in the evolving landscape of diffusion models.

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- AdaptiveFuse: Adaptive Fusion for Diffusion Model Composition (Preparing for submission, first author)
- Large Language Models in Politics and Democracy: A Comprehensive Survey (arXiv:2412.04498 [cs.CL], first author)
- AI Agents in Practice: Harnessing AI for All (UNU Macau AI Conference 2024, speaker)

Patents:

- 一种多扩散模型融合的图像及视频定制化方法、系统和装置, Application No. 202411822988X, (first author)